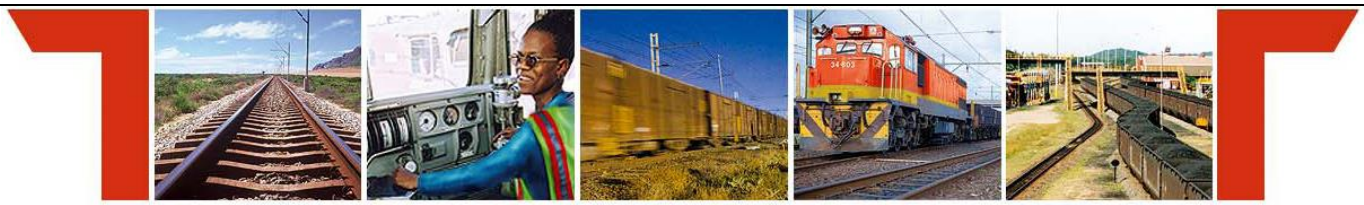


TRANSNET FREIGHT RAIL – TECHNOLOGY MANAGEMENT - TRAIN DESIGN TECHNOLOGIES

PRESCRIBED LOAD TABLES FOR THE RUNNING OF TRANSNET FREIGHT RAIL AIR AND VACUUM BRAKE TRAINS		**Document Number : S.GFB/TES/TE/OI 0272**
Compiling officer:	**A.R. Boy :**	**Revision Number : 18**
1st Technical Approving Officer	**A.R. Boy :**	**Issue date: 2018-01-26**
2nd Technical Approving Officer	**J.M. Mulder :**	**Page Number :**
Approving Safety Officer:	**M. Dlamini :**	**1 of 44**

CONTENTS

		Page
1.	SCOPE	2
2.	INTRODUCTION AND RESTRICTIONS	2
3.	GRADIENT CATEGORIES VS ACTUAL GRADIENTS	5
4.	DIAGRAMS DEPICTING GRADIENT CATEGORIES	6
4.1	DIAGRAM 1 – CAPE TOWN	6
4.2	DIAGRAM 2 – KIMBERLEY	7
4.3	DIAGRAM 3 – PORT ELIZABETH	8
4.4	DIAGRAM 4 – EAST LONDON	9
4.5	DIAGRAM 5 – BLOEMFONTEIN	10
4.6	DIAGRAM 6 – DURBAN	11
4.7	DIAGRAM 7 – JOHANNESBURG	12
4.8	DIAGRAM 8 – PRETORIA	13
4.9	DIAGRAM 9 – RICHARDS BAY	14
4.10	DIAGRAM 10 – SALDANHA	15
5.	LOAD TABLES	16
5.1	GENERAL LOAD TABLE FOR 5E1 LOCOMOTIVES	16
5.2	GENERAL LOAD TABLE FOR 6E LOCOMOTIVES	17
5.3	GENERAL LOAD TABLE FOR 6E1, 16E AND 17E LOCOMOTIVES	18
5.4	GENERAL LOAD TABLE FOR 7E AND 10E LOCOMOTIVES	19
5.5	GENERAL LOAD TABLE FOR 8E LOCOMOTIVES	20
5.6	GENERAL LOAD TABLE FOR 18E LOCOMOTIVES	21
5.7	GENERAL LOAD TABLE FOR 20E LOCOMOTIVES	22
5.8	GENERAL LOAD TABLE FOR 22E LOCOMOTIVES	23
5.9	GENERAL LOAD TABLE FOR 33 DIESEL LOCOMOTIVES	24
5.10	GENERAL LOAD TABLE FOR 34 DIESEL LOCOMOTIVES	25
5.11	GENERAL LOAD TABLE FOR 35 DIESEL LOCOMOTIVES	26
5.12	GENERAL LOAD TABLE FOR 35-800 DIESEL LOCOMOTIVES	27
5.13	GENERAL LOAD TABLE FOR 36 DIESEL LOCOMOTIVES	28
5.14	GENERAL LOAD TABLE FOR 37 AND NEXUS UPGRADED 34 DIESEL LOCOMOTIVES	29
5.15	GENERAL LOAD TABLE FOR 38D/E LOCOMOTIVES	30
5.16	GENERAL LOAD TABLE FOR 39-000 DIESEL LOCOMOTIVES	31
5.17	GENERAL LOAD TABLE FOR 39-200 DIESEL LOCOMOTIVES	32
5.18	GENERAL LOAD TABLE FOR CLASS 43 DIESEL LOCOMOTIVES	33
5.19	GENERAL LOAD TABLE FOR CLASS 44 DIESEL LOCOMOTIVES	34
5.20	GENERAL LOAD TABLE FOR CLASS 44 DIESEL LOCOMOTIVES	35
5.21	GENERAL LOAD TABLE FOR CLASS TAL 2000 DIESEL LOCOMOTIVES	36
5.22	GENERAL LOAD TABLE FOR 'TIME SENSITIVE' LIGHT AIRBRAKE TRAINS	37
5.23	GENERAL LOAD TABLE FOR PASSENGER TRAINS	38

This document is classified as 'Confidential' and as such is authorized and approved for general distribution within Transnet Freight Rail. (The document and/or any information contained herein may not be disclosed outside of TFR without prior permission first being obtained from the Principal Engineer (Train Design Technologies))

1. SCOPE

1.1 The purpose of this document is to facilitate and provide the required prescribed Load Tables for the running of Transnet Freight Rail air and vacuum brake trains.

1.2 This document does not singularly authorize the running of any airbrake train/s. All airbrake trains must be covered by a Technical Train Operating Instruction as specific or unique circumstances for particular trains or routes might well exist. The Load Table/s depicted in a Technical Train Operating Instruction will take precedence over loads as displayed in this document for the running of the specific trains as authorized by such Technical Train Operating Instruction.

1.3 This document also does not authorize the running of any trains conveying damaged, un-roadworthy or unbraked vehicles/locomotives, neither any other train type where special circumstances might exist. A Special Authority for the running of such train types must be obtained from the Technology Management, Train Design Technologies department.

1.4 This document details the Load Tables that should be utilized for such trains and cancels and supersedes documents S.SD/TBF/13/2/1/2 dated 31 August 1995 and any/all earlier revisions of documents with reference S.GFB/TES/TE/OI 0272.

1.5 This revision has been issued whereby provision has been made for the class TAL 2000 diesel locomotives within the load tables provided.

2. INTRODUCTION AND RESTRICTIONS

2.1 The enclosed Load Tables indicate the combinations of locomotives and trainloads that may be operated. The train combinations, maximum tonnage and minimum number of locomotives are reflected in these tables.

2.2 The lowest “Gradient Category” derived from the diagrams provided over the relevant sections must be utilised to determine the load contained in the relevant Load Table.

2.3 The minimum number of locomotives, as indicated in the attached tables, must in all instances be used for tractive effort purposes.

2.4 The absolute maximum number of locomotives in a train consist for airbrake trains is six (6), for vacuum brake trains the maximum is four (4).

2.5 The load of the train includes all wagons and any dead locomotives. If the mass of any dead locomotives can be accommodated within the allowable load, the number of wagons need not be decreased by the equivalent tonnage.

2.6 The electric brakes of active locomotives in a train consist must be engaged to a maximum of four. The exception being classes 6E/6E1, 22E, 23D, 43D, 44D and 45D where the electric brakes of a maximum of three locomotives in the consist must be engaged. This unless otherwise specified or pertinently overruled by a particular Technical Train Operating Instruction, in which case the clause will not be

This document is classified as ‘Confidential’ and as such is authorized and approved for general distribution within Transnet Freight Rail. (The document and/or any information contained herein may not be disclosed outside of TFR without prior permission first being obtained from the Principal Engineer (Train Design Technologies))

relevant and the conditions as stipulated by such document will prevail for the particular train types as authorised accordingly.

2.7 If dead locomotives are hauled as part of the load the number thereof, may not exceed the number of active locomotives in the consist.

2.8 All over-border loads must be compiled, from point of origin, in accordance to the conditions specified in these load tables. This to prevent wagons being unnecessarily detached en-route. Loads are thus not optimal in all cases.

2.9 Axle length restrictions due to station and interloop lengths for the crossing of trains and staging areas must still be adhered to.

2.10 It is essential to specify standard loads for "through" trains, this to eliminate any resultant contradictions.

2.11 Prior to the departure of the train it must be ensured that the number of locomotives as provided for traction purposes are acceptable and prove adequate for the charging of the train brake system. If difficulty is experienced in the charging of the train brake system an examination for potential leaks should be performed. Provided no leaks are identified either additional locomotive/s must be provided to facilitate the charging of the train brake system, or alternatively an appropriate number of wagons should be removed from the train.

2.12 The following clauses are highlighted with reference to document BBD 5249 (Version 1) dated October 2008 and titled '*Alarm Limits for the Measurement of Weighbridges, Wheel Impact Monitors and Skew Bogie Detectors*'.

Overloading.

1. Relevant to both vacuum and airbrake trains the maximum allowable overload permitted for the train to continue is 'One (1) ton X the Number of Wagons on the Train'. Thereafter a sufficient number of wagons must be removed from the train to ensure that the train load is kept within the specified limits.

2. The train must be stopped where the gross wagon mass of any wagon is identified as being 7% or higher for the specific wagon. Any such identified wagon should be removed from the train accordingly.

2.13 The method of load determination for a locomotive consist containing a combination of 34 and 37 class diesel locomotives is as follows, -

1. Determine the number of each specific locomotive type in the consist.
2. Determine the gradient category for the relevant section. (Refer to "Section 4" of this document)
3. Refer to the load tables as contained in "Section 5" of this document, under the relevant locomotive category cross reference the two known variables (gradient category and number of locomotives for each specific locomotive class) for each relevant locomotive class type as contained in the consist to determine the amount of tons that may be hauled over the specific section.

This document is classified as 'Confidential' and as such is authorized and approved for general distribution within Transnet Freight Rail. (The document and/or any information contained herein may not be disclosed outside of TFR without prior permission first being obtained from the Principal Engineer (Train Design Technologies))

4. Add the different determined tonnage amounts together in order to determine the total load the specified combined consist can haul over the relevant section.

It should be noted that this method of calculation is only valid where the balancing speed of the different locomotive classes is the same.

Example

Assume the following, -

1. A four locomotive diesel consist contains,
   - 2 x 34D
   - 2 x 37D

2. The ruling gradient over the section is a 1:35 and the average axle mass loading of the vacuum braked train is 18.5 tons per axle.

As per the above information a 1:35 ruling gradient corresponds to a “gradient category 2”.
(refer to “Section 3” of this document)

Now refer to “Section 5.1” of this document that contains the load table for vacuum brake trains of axle mass loading greater than 10 tons per axle. It can now be established that the loads for 2 x 34 and 2 x 37 class diesel locomotives would be 1140 and 1280 tons respectively. Therefore it can easily be established that the allowable tonnage for the 34 and 37 class diesel locomotives in the consist would be 2420 tons.

Thus the total permissible tonnage for the train would be 2420 tons. (1140 + 1280 = 2420 tons)

2.14 The class 43D and 44D locomotives are solely authorized for MU operation with each other irrespective of which locomotive class is in the ‘lead’ and which is in the ‘trail’ position respectively.

2.15 The class 45D locomotives are currently <u>not</u> authorized to MU with any other class TFR locomotive/s.

2.16 The class 20E locomotives are currently <u>not</u> authorized to MU with any other class TFR locomotive/s.

2.17 The class 21E locomotives are currently <u>not</u> authorized to MU with any other class TFR locomotive/s.

2.18 The classes 22E locomotives are currently solely authorized to MU with diesel locomotive/s.

2.19 The class 35-800D locomotives are <u>not</u> authorized to MU with any other class TFR diesel locomotives as it is only possible to couple the brake pipe through. It must additionally be noted that the brake pipe needs to be coupled straight across and <u>not</u> under the coupler as is the current TFR regime, as this leads to the pipes uncoupling when negotiating a cross over or tight curve.

This document is classified as ‘Confidential’ and as such is authorized and approved for general distribution within Transnet Freight Rail. (The document and/or any information contained herein may not be disclosed outside of TFR without prior permission first being obtained from the Principal Engineer (Train Design Technologies))

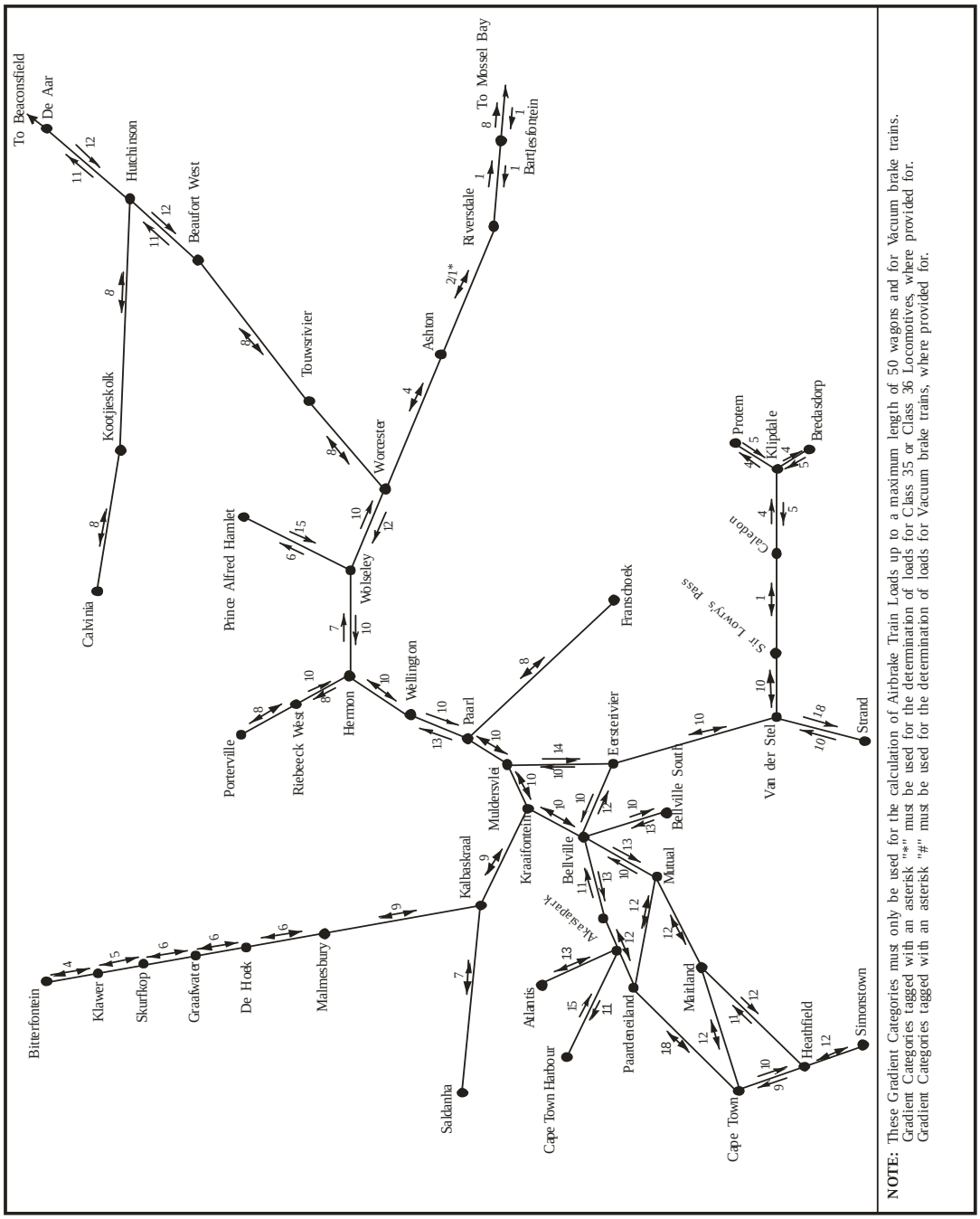
3. GRADIENT CATEGORIES VS ACTUAL GRADIENTS

Gradient Category	=	Actual Gradient
1	=	1 : 30 → 1 : 34
2	=	1 : 35 → 1 : 39
3	=	1 : 40 → 1 : 44
4	=	1 : 45 → 1 : 49
5	=	1 : 50 → 1 : 54
6	=	1 : 55 → 1 : 59
7	=	1 : 60 → 1 : 65
8	=	1 : 66 → 1 : 69
9	=	1 : 70 → 1 : 79
10	=	1 : 80 → 1 : 89
11	=	1 : 90 → 1 : 99
12	=	1 : 100 → 1 : 109
13	=	1 : 110 → 1 : 119
14	=	1 : 120 → 1 : 149
15	=	1 : 150 → 1 : 179
16	=	1 : 180 → 1 : 249
17	=	1 : 250 → 1 : 279
18	=	1 : 280 →

This document is classified as 'Confidential' and as such is authorized and approved for general distribution within Transnet Freight Rail. (The document and/or any information contained herein may not be disclosed outside of TFR without prior permission first being obtained from the Principal Engineer (Train Design Technologies))

4. DIAGRAMS DEPICTING GRADIENT CATEGORIES

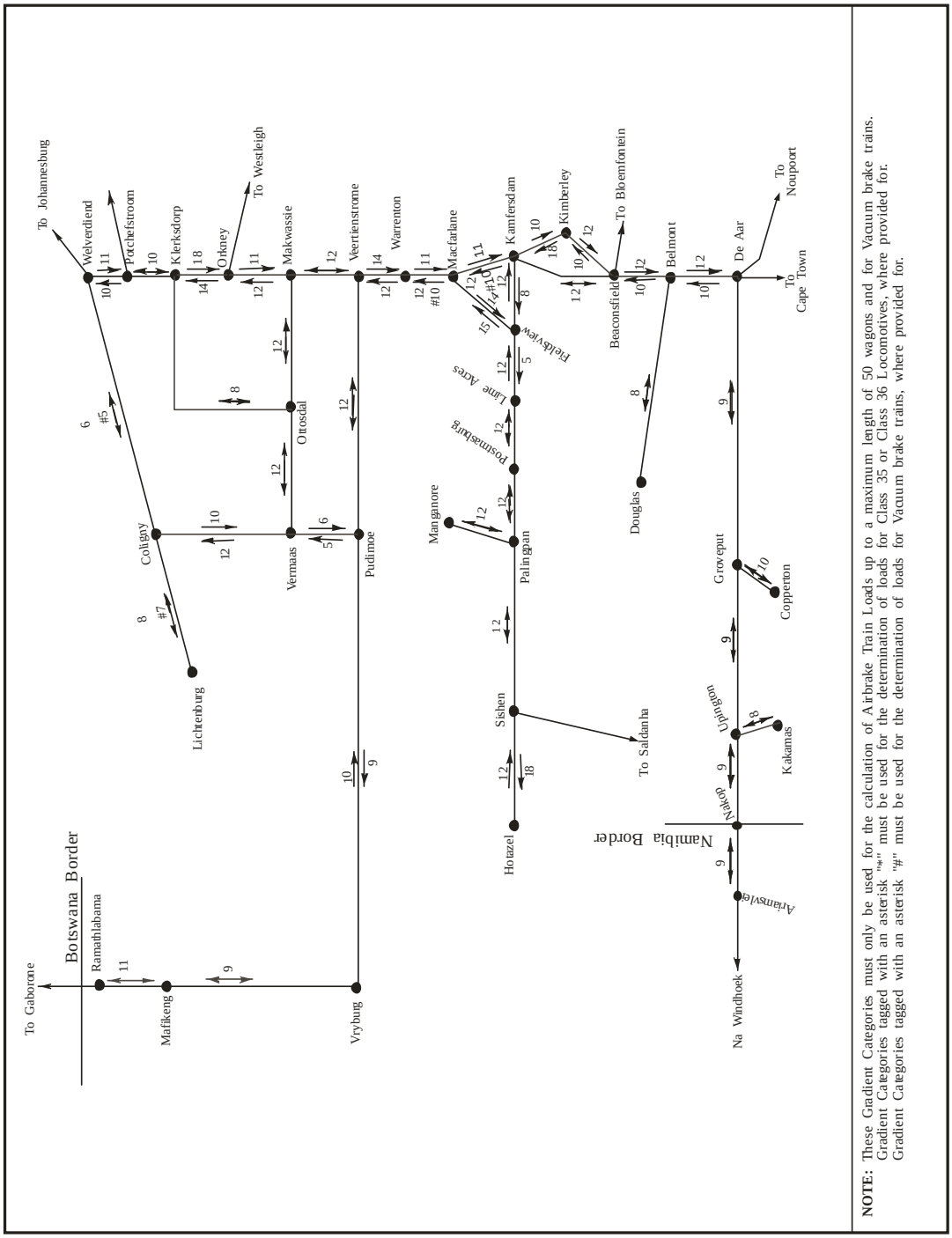
4.1 DIAGRAM 1 – CAPE TOWN



NOTE: These Gradient Categories must only be used for the calculation of Airbrake Train Loads up to a maximum length of 50 wagons and for Vacuum brake trains.
Gradient Categories tagged with an asterisk "*" must be used for the determination of loads for Class 35 or Class 36 Locomotives, where provided for.
Gradient Categories tagged with an asterisk "#" must be used for the determination of loads for Vacuum brake trains, where provided for.

This document is classified as 'Confidential' and as such is authorized and approved for general distribution within Transnet Freight Rail. (The document and/or any information contained herein may not be disclosed outside of TFR without prior permission first being obtained from the Principal Engineer (Train Design Technologies))

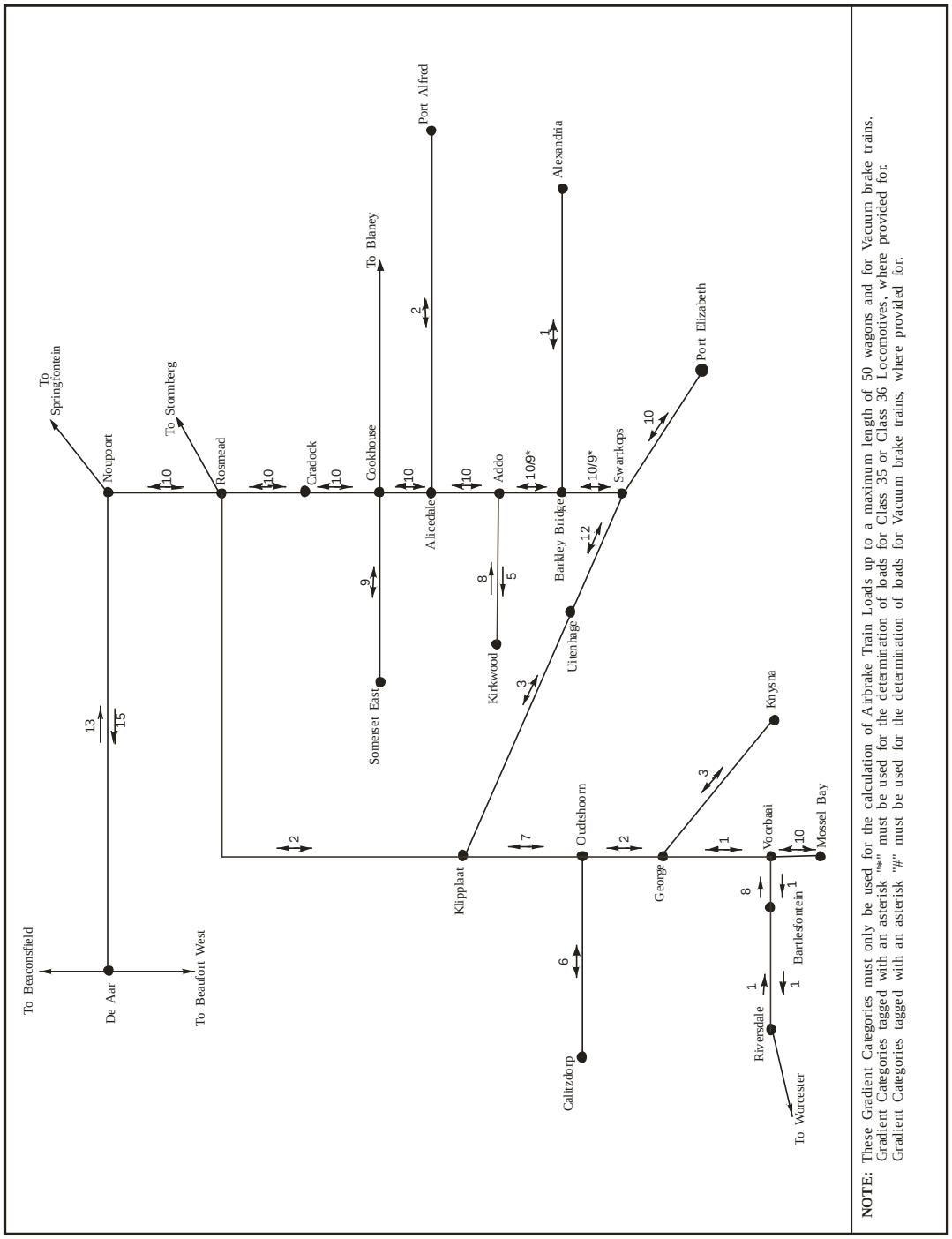
4.2 DIAGRAM 2 - KIMBERLEY



NOTE: These Gradient Categories must only be used for the calculation of Airbrake Train Loads up to a maximum length of 50 wagons and for Vacuum brake trains. Gradient Categories tagged with an asterisk "*" must be used for the determination of loads for Class 35 or Class 36 Locomotives, where provided for. Gradient Categories tagged with an asterisk "#" must be used for the determination of loads for Vacuum brake trains, where provided for.

This document is classified as 'Confidential' and as such is authorized and approved for general distribution within Transnet Freight Rail. (The document and/or any information contained herein may not be disclosed outside of TFR without prior permission first being obtained from the Principal Engineer (Train Design Technologies))

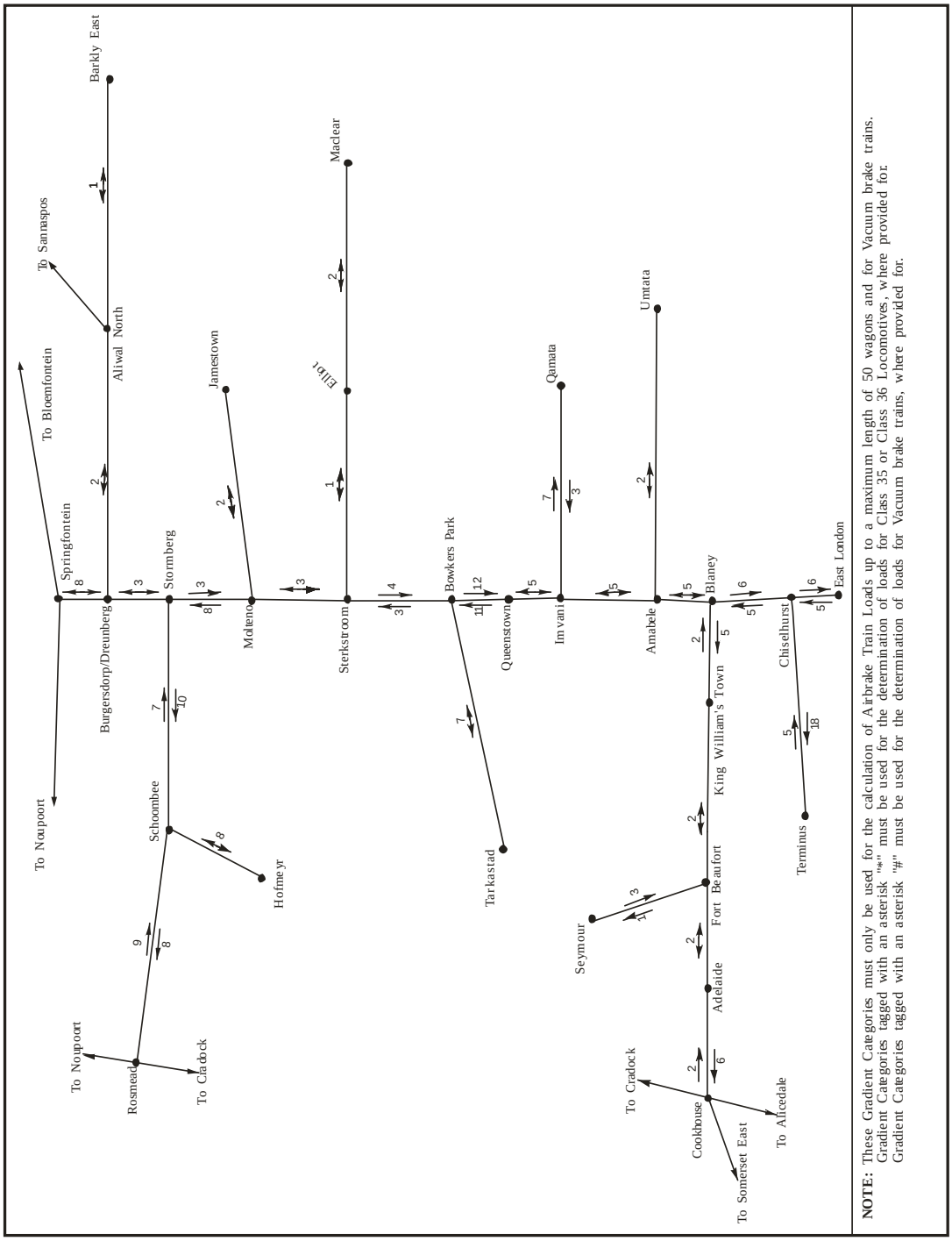
4.3 DIAGRAM 3 – PORT ELIZABETH



NOTE: These Gradient Categories must only be used for the calculation of Airbrake Train Loads up to a maximum length of 50 wagons and for Vacuum brake trains.
Gradient Categories tagged with an asterisk "*" must be used for the determination of loads for Class 35 or Class 36 Locomotives, where provided for.
Gradient Categories tagged with an asterisk "#" must be used for the determination of loads for Vacuum brake trains, where provided for.

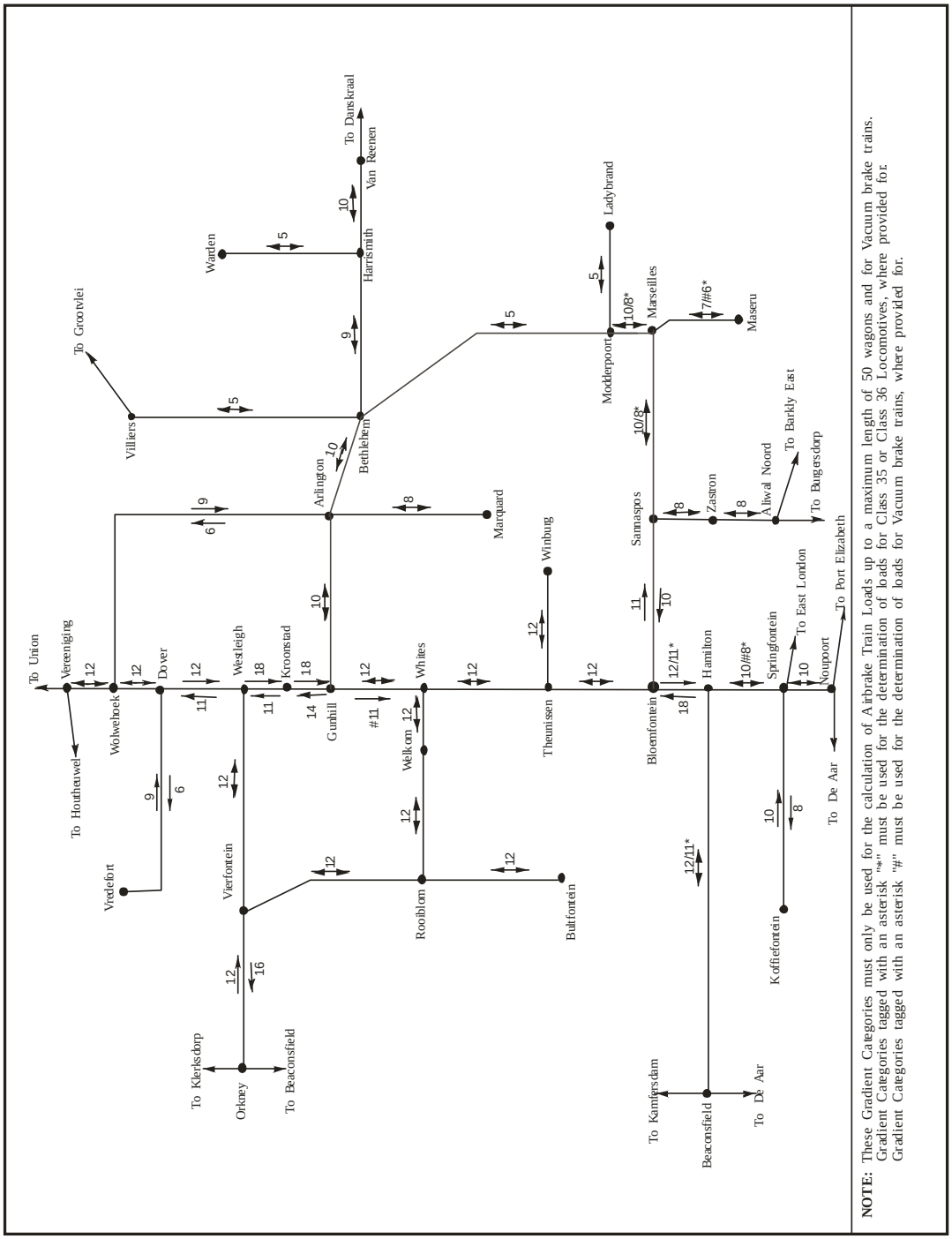
This document is classified as 'Confidential' and as such is authorized and approved for general distribution within Transnet Freight Rail. (The document and/or any information contained herein may not be disclosed outside of TFR without prior permission first being obtained from the Principal Engineer (Train Design Technologies))

4.4 DIAGRAM 4 – EAST LONDON



This document is classified as 'Confidential' and as such is authorized and approved for general distribution within Transnet Freight Rail. (The document and/or any information contained herein may not be disclosed outside of TFR without prior permission first being obtained from the Principal Engineer (Train Design Technologies))

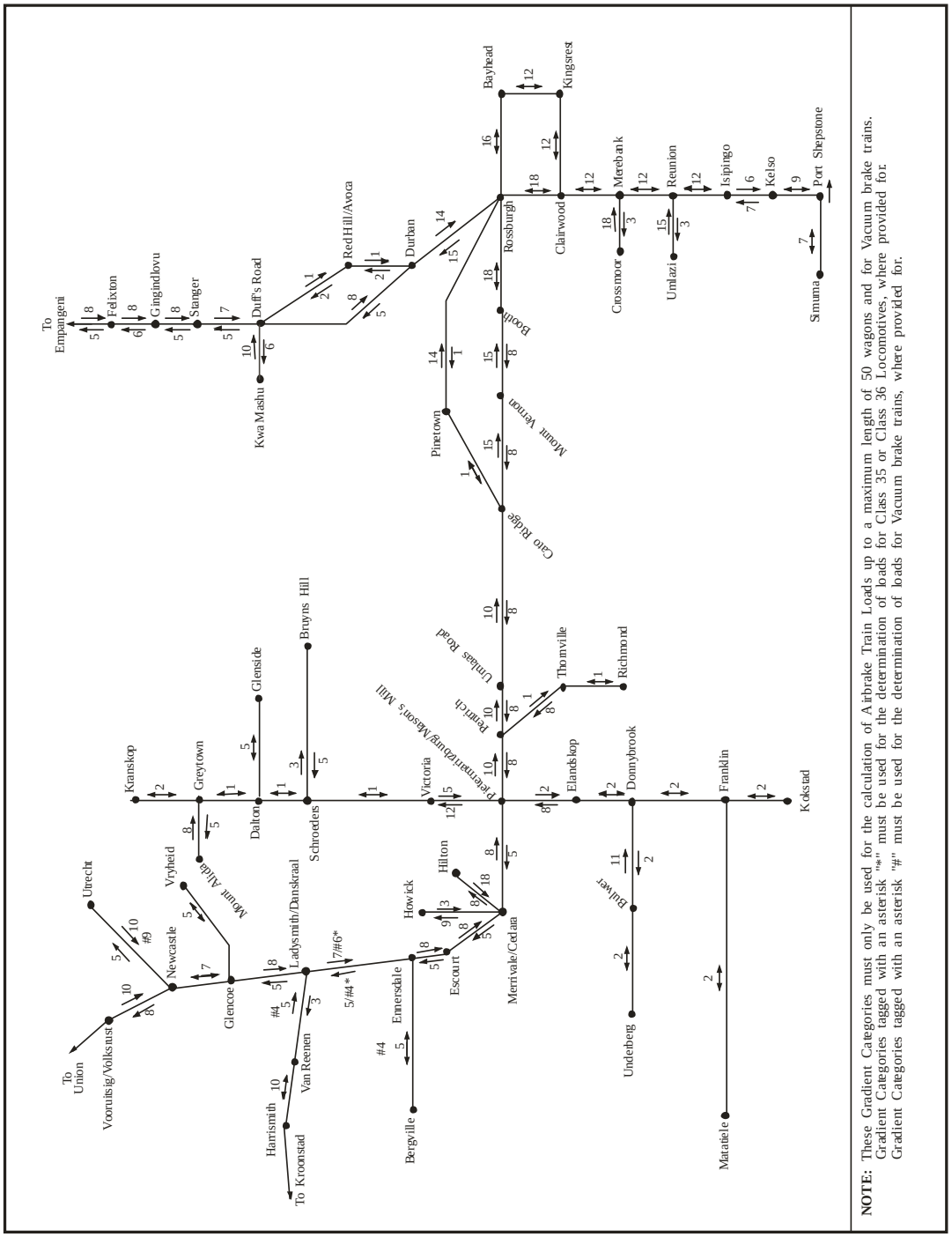
4.5 DIAGRAM 5 - BLOEMFONTEIN



NOTE: These Gradient Categories must only be used for the calculation of Airbrake Train Loads up to a maximum length of 50 wagons and for Vacuum brake trains. Gradient Categories tagged with an asterisk "*" must be used for the determination of loads for Class 35 or Class 36 Locomotives, where provided for. Gradient Categories tagged with an asterisk "#" must be used for the determination of loads for Vacuum brake trains, where provided for.

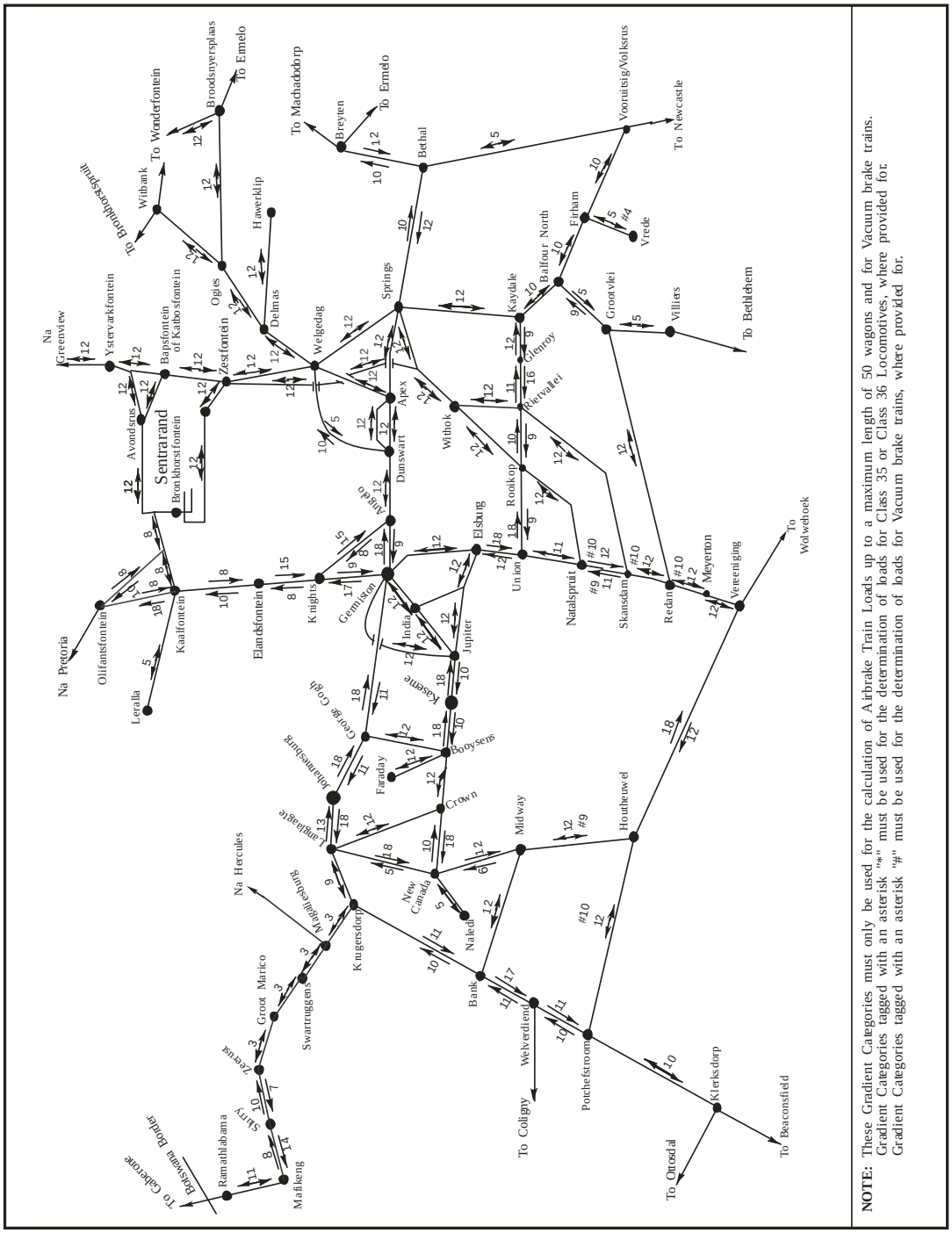
This document is classified as 'Confidential' and as such is authorized and approved for general distribution within Transnet Freight Rail. (The document and/or any information contained herein may not be disclosed outside of TFR without prior permission first being obtained from the Principal Engineer (Train Design Technologies))

4.6 DIAGRAM 6 - DURBAN



This document is classified as 'Confidential' and as such is authorized and approved for general distribution within Transnet Freight Rail. (The document and/or any information contained herein may not be disclosed outside of TFR without prior permission first being obtained from the Principal Engineer (Train Design Technologies))

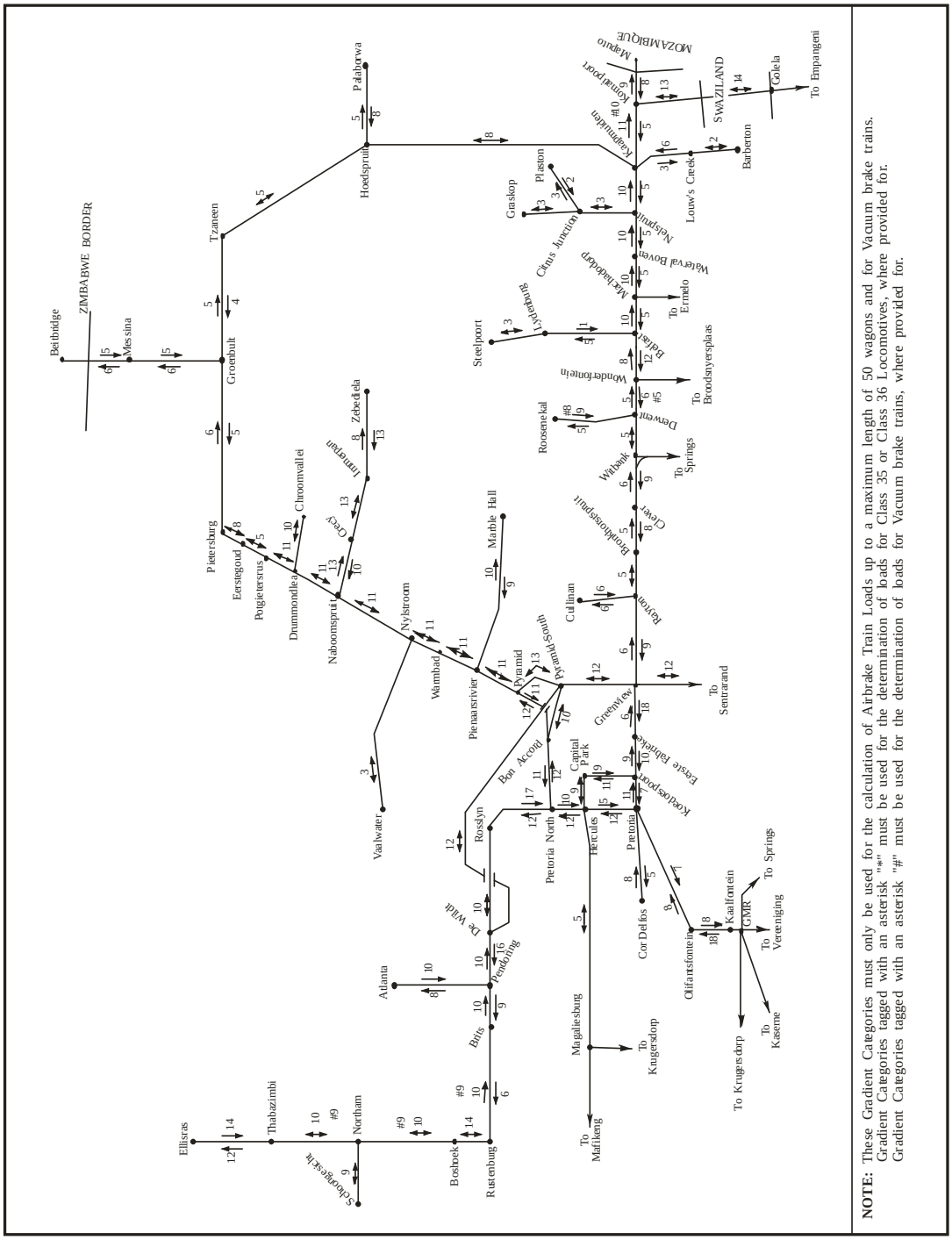
4.7 DIAGRAM 7 - JOHANNESBURG



NOTE: These Gradient Categories must only be used for the calculation of Airbrake Train Loads up to a maximum length of 50 wagons and for Vacuum brake trains.
Gradient Categories tagged with an asterisk "*" must be used for the determination of loads for Class 35 or Class 36 Locomotives, where provided for.
Gradient Categories tagged with an asterisk "#" must be used for the determination of loads for Vacuum brake trains, where provided for.

This document is classified as 'Confidential' and as such is authorized and approved for general distribution within Transnet Freight Rail. (The document and/or any information contained herein may not be disclosed outside of TFR without prior permission first being obtained from the Principal Engineer (Train Design Technologies))

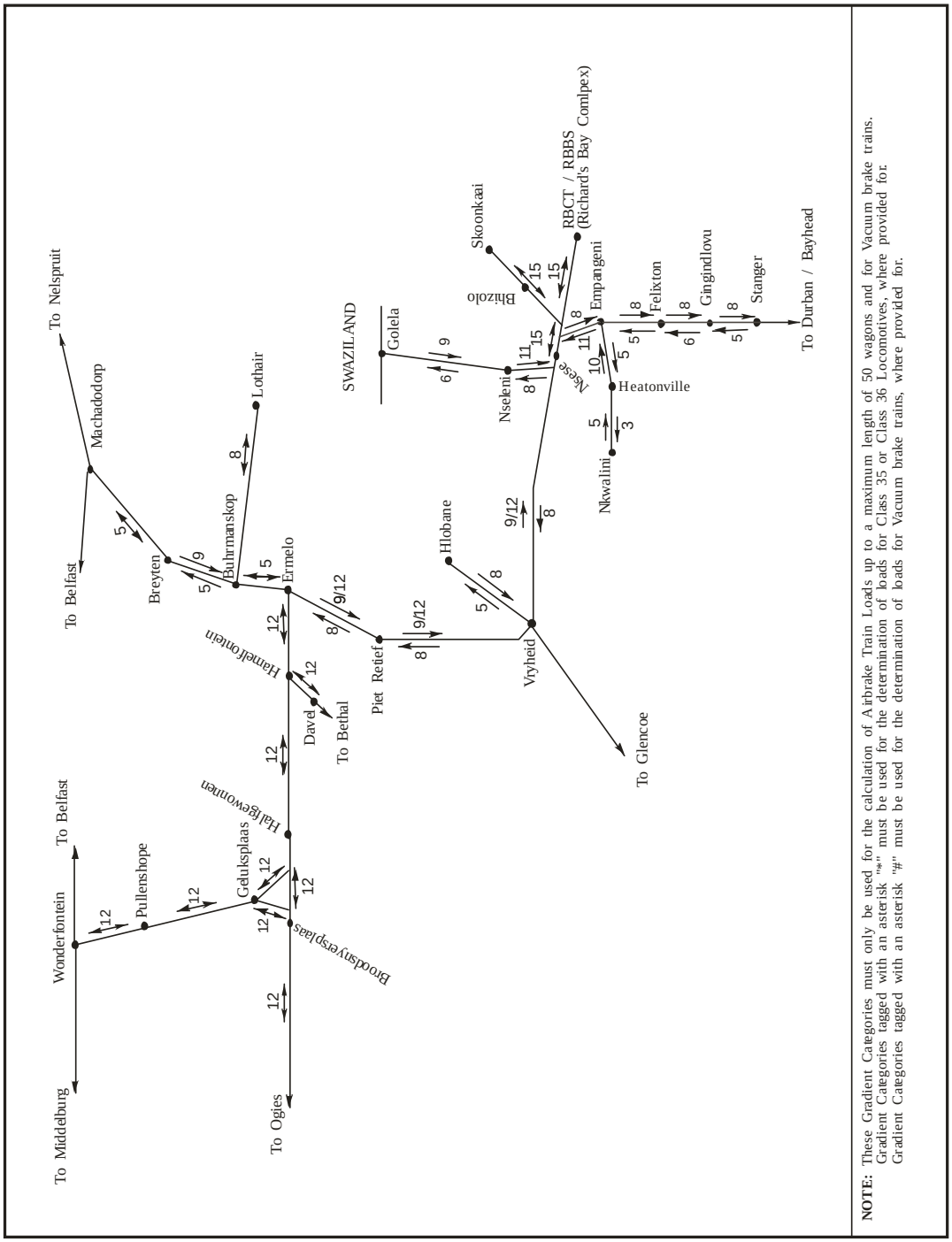
4.8 DIAGRAM 8 - PRETORIA



NOTE: These Gradient Categories must only be used for the calculation of Airbrake Train Loads up to a maximum length of 50 wagons and for Vacuum brake trains. Gradient Categories tagged with an asterisk "*" must be used for the determination of loads for Class 35 or Class 36 Locomotives, where provided for. Gradient Categories tagged with an asterisk "#" must be used for the determination of loads for Vacuum brake trains, where provided for.

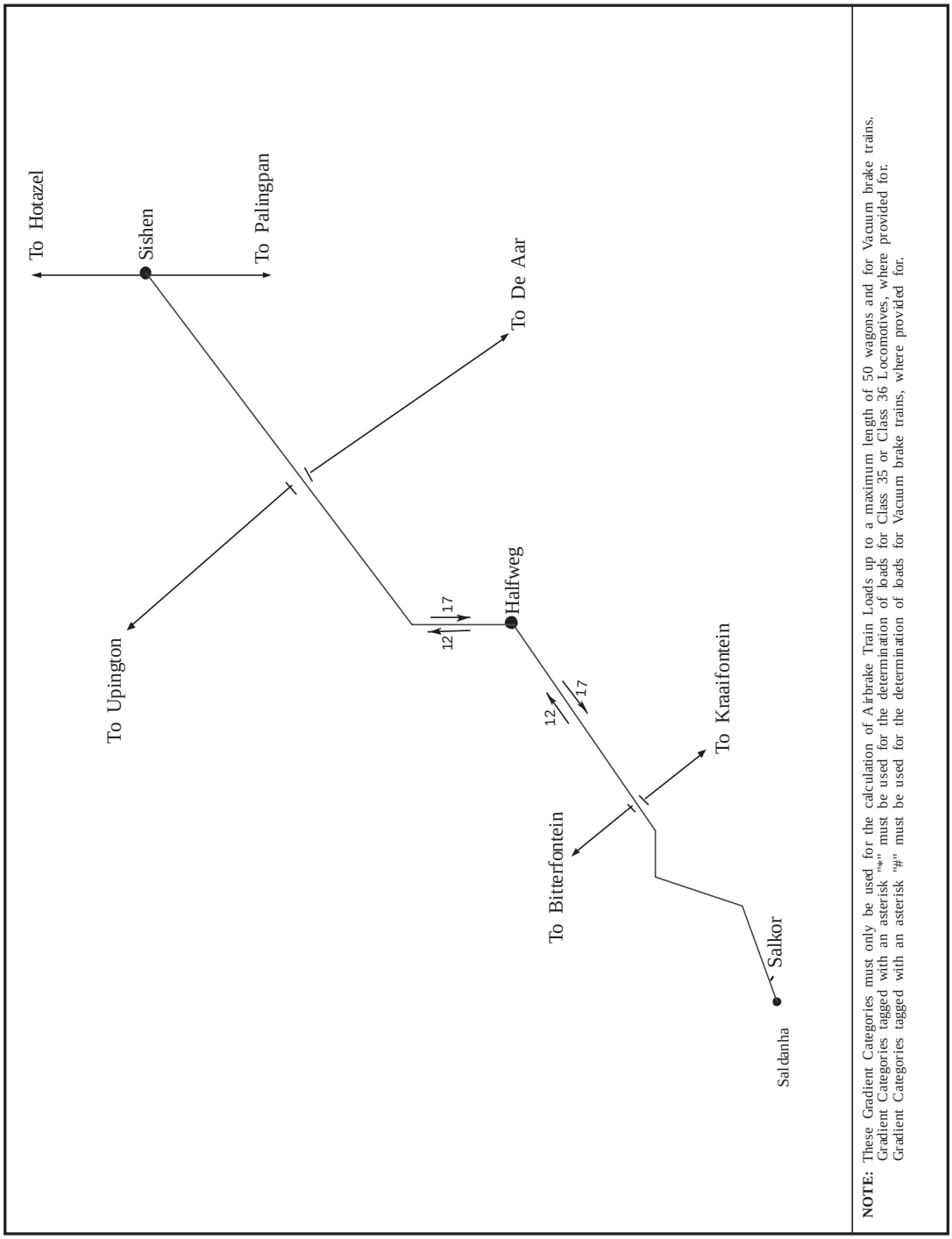
This document is classified as 'Confidential' and as such is authorized and approved for general distribution within Transnet Freight Rail. (The document and/or any information contained herein may not be disclosed outside of TFR without prior permission first being obtained from the Principal Engineer (Train Design Technologies))

4.9 DIAGRAM 9 – RICHARDS BAY



This document is classified as 'Confidential' and as such is authorized and approved for general distribution within Transnet Freight Rail. (The document and/or any information contained herein may not be disclosed outside of TFR without prior permission first being obtained from the Principal Engineer (Train Design Technologies))

4.10 DIAGRAM 10 - SALDANHA



NOTE: These Gradient Categories must only be used for the calculation of Airbrake Train Loads up to a maximum length of 50 wagons and for Vacuum brake trains. Gradient Categories tagged with an asterisk "*" must be used for the determination of loads for Class 35 or Class 36 Locomotives, where provided for. Gradient Categories tagged with an asterisk "#" must be used for the determination of loads for Vacuum brake trains, where provided for.

This document is classified as 'Confidential' and as such is authorized and approved for general distribution within Transnet Freight Rail. (The document and/or any information contained herein may not be disclosed outside of TFR without prior permission first being obtained from the Principal Engineer (Train Design Technologies))

5. LOAD TABLES

5.1 GENERAL LOAD TABLES FOR 5E1 LOCOMOTIVES

LOAD TABLES FOR 5E1 LOCOMOTIVES
AIRBRAKE - 50 WAGON TRAIN LENGTH MAXIMUM
VACUUM BRAKE - 40 WAGON TRAIN LENGTH MAXIMUM

TONNAGE RATINGS BY LINE PER LOCOMOTIVE

5E1 ELECTRIC LOCOMOTIVES

VACUUM BRAKED TRAINS ≤ 10 TONS PER AXLE							
CLASS : 5E1							
		(NUMBER)					
GC	GRAD.	1	2	3	4	5	6
1	30	250	490	740	980	-	-
2	35	290	590	880	1170		
3	40	340	680	1020	1360		
4	45	390	770	1160	1540		
5	50	430	860	1290	1600		
6	55	470	940	1410	1600		
7	60	510	1020	1530	1600		
8	66	560	1120	1600			
9	70	590	1180	1600			
10	80	660	1320	1600			
11	90	730	1460	1600			
12	100	790	1580	1600			
13	110	850	1600				
14	120	910	1600				
15	150	1060	1600				
16	200	1270	1600				
17	250	1440	1600				
18	280	1520	1600				

VACUUM BRAKED TRAINS > 10 TONS PER AXLE						
CLASS : 5E1						
	(NUMBER)					
GC	1	2	3	4	5	6
1	260	520	780	1040	-	-
2	310	630	940	1260		
3	370	730	1100	1470		
4	420	840	1250	1670		
5	470	930	1400	1870		
6	520	1030	1550	2060		
7	560	1130	1690	2260		
8	620	1240	1860	2480		
9	660	1310	1970	2630		
10	740	1490	2230	2960		
11	830	1660	2490	2960		
12	910	1820	2730	2960		
13	990	1980	2960			
14	1070	2130	2960			
15	1280	2550	2960			
16	1580	2960				
17	1840	2960				
18	1970	2960				

AIRBRAKE TRAINS >2 to 7 t/a						
CLASS : 5E1						
	(NUMBER)					
GC	1	2	3	4	5	6
1	210	420	640	850	-	-
2	250	500	750	1000		
3	290	570	860	1150		
4	320	640	960	1280		
5	350	710	1060	1400		
6	380	770	1150	1400		
7	410	820	1240	1400		
8	440	890	1330	1400		
9	460	930	1390	1400		
10	510	1020	1400			
11	550	1110	1400			
12	590	1190	1400			
13	630	1260	1400			
14	660	1320	1400			
15	740	1400				
16	850	1400				
17	920	1400				
18	960	1400				

AIRBRAKE TRAINS >7 to 12.5 t/a						
CLASS : 5E1						
	(NUMBER)					
GC	1	2	3	4	5	6
1	250	510	760	1020	-	-
2	300	610	910	1220		
3	350	710	1060	1420		
4	400	800	1210	1610		
5	450	900	1350	1800		
6	490	990	1480	1980		
7	540	1080	1620	2160		
8	590	1180	1770	2360		
9	620	1250	1870	2500		
10	700	1410	2110	2500		
11	780	1560	2340	2500		
12	850	1710	2500			
13	920	1850	2500			
14	990	1980	2500			
15	1170	2340	2500			
16	1430	2500				
17	1630	2500				
18	1740	2500				

5E1 ELECTRIC LOCOMOTIVES

AIRBRAKE TRAINS >12.5 to 17 t/a							
CLASS : 5E1							
		(Number)					
GC	GRAD.	1	2	3	4	5	6
1	30	260	530	790	1060	-	-
2	35	320	640	960	1280		
3	40	370	750	1120	1490		
4	45	430	850	1280	1700		
5	50	480	950	1430	1910		
6	55	530	1060	1580	2110		
7	60	580	1150	1730	2310		
8	66	640	1270	1910	2540		
9	70	670	1350	2020	2690		
10	80	770	1530	2300	3070		
11	90	860	1710	2570	3400		
12	100	940	1890	2830	3400		
13	110	1030	2050	3080	3400		
14	120	1110	2220	3320	3400		
15	150	1340	2670	3400			
16	200	1670	3340	3400			
17	250	1960	3400				
18	280	2110	3400				

AIRBRAKE TRAINS >17 to 19 t/a						
CLASS : 5E1						
	(Number)					
GC	1	2	3	4	5	6
1	270	540	810	1080	-	-
2	330	650	980	1300		
3	380	760	1140	1520		
4	430	870	1300	1740		
5	490	980	1460	1950		
6	540	1080	1620	2160		
7	590	1190	1780	2370		
8	650	1310	1960	2620		
9	690	1390	2080	2780		
10	790	1590	2380	3170		
11	890	1780	2660	3550		
12	980	1960	2940	3800		
13	1070	2140	3210	3800		
14	1160	2320	3480	3800		
15	1410	2820	3800			
16	1780	3570	3800			
17	2110	3800				
18	2290	3800				

AIRBRAKE TRAINS >19 to 20 t/a						
CLASS : 5E1						
	(Number)					
GC	1	2	3	4	5	6
1	270	540	810	1080	-	-
2	330	650	980	1310		
3	380	770	1150	1530		
4	440	880	1310	1750		
5	490	980	1480	1970		
6	550	1090	1640	2180		
7	600	1200	1790	2390		
8	660	1320	1980	2640		
9	700	1400	2100	2800		
10	800	1600	2400	3200		
11	900	1800	2700	3590		
12	990	1990	2980	3970		
13	1090	2170	3260	4000		
14	1180	2350	3530	4000		
15	1430	2870	4000			
16	1820	3640	4000			
17	2170	4000				
18	2360	4000				

MOTOR CAR TRAINS, 8 to 12 t/a #						
CLASS : 5E1						
	(Number)					
GC	1	2	3	4	5	6
1	190	380	560	750	-	-
2	230	460	690	920		
3	270	540	810	1080		
4	310	620	930	1240		
5	350	700	1040	1390		
6	390	770	1160	1540		
7	420	840	1270	1690		
8	460	930	1390	1860		
9	490	980	1480	1970		
10	560	1120	1680	2240		
11	620	1250	1870	2400		
12	680	1370	2050	2400		
13	740	1480	2230	2400		
14	800	1600	2390	2400		
15	950	1900	2400			
16	1170	2340	2400			
17	1350	2400				
18	1450	2400				

RELEVANT TO VACUUM BRAKED TRAINS THE FOLLOWING RESTRICTIONS MAY NOT BE EXCEEDED:

a. Maximum train speed of 60 km/h.
b. Maximum average axle load per train of 18.5 ton per axle.
c. Maximum axle load per truck per axle of 20 ton.
d. Maximum train length of 40 trucks (160 axles).
e. Maximum train length on specific line as restricted in WTB.
f. Maximum axle load restriction on specific line.
g. Maximum total load of 2560 ton on category 1 lines where SASKOP-couplers are used.
h. Maximum number of locomotives in a consist is four (4).

Notes :

a. 'GC' Means Gradient Category.
b. Refer to point 5.9 below for Load Tables relevant to Passenger Trains.

This document is classified as 'Confidential' and as such is authorized and approved for general distribution within Transnet Freight Rail. (The document and/or any information contained herein may not be disclosed outside of TFR without prior permission first being obtained from the Principal Engineer (Train Design Technologies))

5.2 GENERAL LOAD TABLES FOR 6E LOCOMOTIVES

LOAD TABLES FOR 6E LOCOMOTIVES
AIRBRAKE - 50 WAGON TRAIN LENGTH MAXIMUM
VACUUM BRAKE - 40 WAGON TRAIN LENGTH MAXIMUM

TONNAGE RATINGS BY LINE PER LOCOMOTIVE

6E ELECTRIC LOCOMOTIVES

VACUUM BRAKED TRAINS ≤ 10 TONS PER AXLE								VACUUM BRAKED TRAINS > 10 TONS PER AXLE							AIRBRAKE TRAINS >2 to 7 t/a							AIRBRAKE TRAINS >7 to 12.5 t/a						
CLASS : 6E								CLASS : 6E							CLASS : 6E							CLASS : 6E						
		(NUMBER)							(NUMBER)							(NUMBER)							(NUMBER)					
GC	GRAD.	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6
1	30	320	640	960	1280	-	-	1	340	680	1020	1350	-	-	1	280	560	840	1110	1390	-	1	330	660	990	1320	1650	1980
2	35	380	760	1140	1510			2	400	810	1210	1620			2	330	650	980	1300			2	390	780	1180	1570	1960	2350
3	40	440	870	1310	1600			3	470	940	1400	1870			3	370	740	1110	1400			3	450	910	1360	1810	2260	2500
4	45	490	980	1470	1600			4	530	1060	1590	2120			4	410	820	1240	1400			4	510	1020	1540	2050	2500	
5	50	540	1090	1600				5	590	1180	1780	2370			5	450	900	1350				5	570	1140	1710	2280	2500	
6	55	600	1190	1600				6	650	1300	1960	2610			6	490	980	1400				6	630	1250	1880	2500		
7	60	650	1290	1600				7	710	1420	2130	2840			7	520	1050	1400				7	680	1360	2040	2500		
8	66	700	1410	1600				8	780	1560	2340	2960			8	560	1130	1400				8	740	1490	2230	2500		
9	70	740	1480	1600				9	820	1650	2470	2960			9	590	1180	1400				9	780	1570	2350			
10	80	830	1600					10	930	1870	2800				10	650	1290	1400				10	880	1770	2500			
11	90	910	1600					11	1040	2080	2960				11	700	1400					11	980	1960	2500			
12	100	990	1600					12	1140	2280	2960				12	750	1400					12	1070	2130	2500			
13	110	1070	1600					13	1230	2470	2960				13	790	1400					13	1150	2310	2500			
14	120	1140	1600					14	1330	2650	2960				14	830	1400					14	1230	2470				
15	150	1320	1600					15	1590	2960					15	930	1400					15	1460	2500				
16	200	1580						16	1960	2960					16	1060	1400					16	1770	2500				
17	250	1600						17	2280	2960					17	1150	1400					17	2030	2500				
18	280	1600						18	2440	2960					18	1200	1400					18	2160	2500				

6E ELECTRIC LOCOMOTIVES

AIRBRAKE TRAINS >12.5 to 17 t/a								AIRBRAKE TRAINS >17 to 19 t/a							AIRBRAKE TRAINS >19 to 20 t/a							MOTOR CAR TRAINS, 8 to 12 t/a #						
CLASS : 6E								CLASS : 6E							CLASS : 6E							CLASS : 6E						
		(Number)							(Number)							(Number)							(Number)					
GC	GRAD.	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6
1	30	340	690	1030	1370	1710	2060	1	350	700	1040	1390	1740	2090	1	350	700	1050	1400	1750	2100	1	250	500	750	990	1240	1490
2	35	410	820	1230	1640	2050	2460	2	420	830	1250	1670	2090	2500	2	420	840	1260	1680	2100	2520	2	300	600	900	1200	1500	1800
3	40	480	950	1430	1900	2380	2860	3	490	970	1460	1940	2430	2910	3	490	980	1460	1950	2440	2930	3	350	700	1050	1400	1750	2100
4	45	540	1080	1620	2160	2700	3240	4	550	1100	1660	2210	2760	3310	4	560	1110	1670	2220	2780	3330	4	400	800	1190	1590	1990	2390
5	50	600	1210	1810	2410	3020	3400	5	620	1240	1850	2470	3090	3710	5	620	1240	1870	2490	3110	3730	5	450	890	1340	1780	2230	
6	55	670	1330	2000	2660	3330		6	680	1360	2050	2730	3410	3800	6	690	1370	2060	2750	3440	4000	6	490	980	1470	1960	2400	
7	60	730	1450	2180	2910	3400		7	750	1490	2240	2980	3730		7	750	1500	2260	3010	3760		7	540	1070	1610	2140	2400	
8	66	800	1600	2400	3190			8	820	1640	2460	3290	3800		8	830	1660	2490	3310	4000		8	590	1180	1760	2350		
9	70	850	1690	2540	3380			9	870	1740	2610	3480	3800		9	880	1760	2640	3520	4000		9	620	1240	1870	2400		
10	80	960	1920	2880	3400			10	990	1980	2980	3800			10	1000	2000	3010	4000			10	700	1410	2110	2400		
11	90	1070	2140	3210				11	1110	2220	3330	3800			11	1120	2240	3370	4000			11	780	1570	2350			
12	100	1180	2350	3400				12	1220	2450	3670				12	1240	2480	3720	4000			12	860	1720	2400			
13	110	1280	2560	3400				13	1330	2670	3800				13	1350	2710	4000				13	930	1860	2400			
14	120	1380	2760	3400				14	1440	2890	3800				14	1460	2930	4000				14	1000	2000	2400			
15	150	1660	3320					15	1750	3500	3800				15	1780	3560	4000				15	1190	2380				
16	200	2070	3400					16	2210	3800					16	2260	4000					16	1460	2400				
17	250	2420	3400					17	2620	3800					17	2680	4000					17	1680	2400				
18	280	2620	3400					18	2840	3800					18	2920	4000					18	1800	2400				

RELEVANT TO VACUUM BRAKED TRAINS THE FOLLOWING RESTRICTIONS MAY NOT BE EXCEEDED:

a. Maximum train speed of 60 km/h.
b. Maximum average axle load per train of 18.5 ton per axle.
c. Maximum axle load per truck per axle of 20 ton.
d. Maximum train length of 40 trucks (160 axles).
e. Maximum train length on specific line as restricted in WTB.
f. Maximum axle load restriction on specific line.
g. Maximum total load of 2560 ton on category 1 lines where SASKOP-couplers are used.
h. Maximum number of locomotives in a consist is four (4).

Notes :

a. 'GC' Means Gradient Category.
b. Refer to point 5.9 below for Load Tables relevant to Passenger Trains.

This document is classified as 'Confidential' and as such is authorized and approved for general distribution within Transnet Freight Rail. (The document and/or any information contained herein may not be disclosed outside of TFR without prior permission first being obtained from the Principal Engineer (Train Design Technologies))

5.3 GENERAL LOAD TABLES FOR 6E1/16E/17E LOCOMOTIVES

LOAD TABLES FOR 6E1/16E/17E LOCOMOTIVES
AIRBRAKE - 50 WAGON TRAIN LENGTH MAXIMUM
VACUUM BRAKE - 40 WAGON TRAIN LENGTH MAXIMUM

TONNAGE RATINGS BY LINE PER LOCOMOTIVE

6E1/16E/17E ELECTRIC LOCOMOTIVES

VACUUM BRAKED TRAINS ≤ 10 TONS PER AXLE								VACUUM BRAKED TRAINS > 10 TONS PER AXLE								AIRBRAKE TRAINS >2 to 7 t/a								AIRBRAKE TRAINS >7 to 12.5 t/a							
CLASS : 6E1/16E/17E								CLASS : 6E1/16E/17E								CLASS : 6E1/16E/17E								CLASS : 6E1/16E/17E							
		(NUMBER)							(NUMBER)								(NUMBER)								(NUMBER)						
GC	GRAD.	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6
1	30	380	760	1140	1530	-	-	1	400	810	1210	1620	-	-	1	330	670	1000	1340	1400	-	1	390	790	1180	1570	1970	2360	1	390	790	1180	1570	1970	2360
2	35	450	900	1350	1600			2	480	960	1440	1920			2	390	780	1170	1400			2	470	930	1400	1860	2330	2500	2	470	930	1400	1860	2330	2500
3	40	520	1030	1550	1600			3	550	1110	1660	2210			3	440	880	1320	1400			3	540	1070	1610	2140	2500		3	540	1070	1610	2140	2500	
4	45	580	1160	1600				4	630	1250	1880	2500			4	490	980	1400				4	600	1210	1810	2420	2500		4	600	1210	1810	2420	2500	
5	50	640	1280	1600				5	700	1390	2090	2790			5	530	1070	1400				5	670	1340	2010	2500			5	670	1340	2010	2500		
6	55	700	1400	1600				6	770	1530	2300	2960			6	580	1150	1400				6	740	1470	2210	2500			6	740	1470	2210	2500		
7	60	760	1520	1600				7	830	1670	2500	2960			7	620	1240	1400				7	800	1600	2390	2500			7	800	1600	2390	2500		
8	66	820	1600					8	910	1820	2740	2960			8	660	1330	1400				8	870	1740	2500				8	870	1740	2500			
9	70	870	1600					9	960	1930	2890	2960			9	690	1400					9	920	1840	2500				9	920	1840	2500			
10	80	970	1600					10	1090	2180	2960				10	760	1400					10	1030	2070	2500				10	1030	2070	2500			
11	90	1070	1600					11	1210	2420	2960				11	820	1400					11	1140	2280	2500				11	1140	2280	2500			
12	100	1160	1600					12	1330	2650	2960				12	880	1400					12	1240	2490	2500				12	1240	2490	2500			
13	110	1240	1600					13	1440	2880	2960				13	930	1400					13	1340	2500					13	1340	2500				
14	120	1320	1600					14	1540	2960					14	970	1400					14	1440	2500					14	1440	2500				
15	150	1540	1600					15	1840	2960					15	1090	1400					15	1690	2500					15	1690	2500				
16	200	1600						16	2270	2960					16	1240	1400					16	2060	2500					16	2060	2500				
17	250	1600						17	2640	2960					17	1350	1400					17	2350	2500					17	2350	2500				
18	280	1600						18	2830	2960					18	1400						18	2500						18	2500					

6E1/16E/17E ELECTRIC LOCOMOTIVES

AIRBRAKE TRAINS >12.5 to 17 t/a								AIRBRAKE TRAINS >17 to 19 t/a								AIRBRAKE TRAINS >19 to 20 t/a								MOTOR CAR TRAINS, 8 to 12 t/a #							
CLASS : 6E1/16E/17E								CLASS : 6E1/16E/17E								CLASS : 6E1/16E/17E								CLASS : 6E1/16E/17E							
		(Number)							(Number)								(Number)								(Number)						
GC	GRAD.	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6
1	30	410	820	1230	1640	2050	2460	1	420	830	1250	1660	2080	2490	1	420	830	1250	1670	2090	2500	1	300	600	900	1200	1500	1800	1	300	600	900	1200	1500	1800
2	35	490	970	1460	1950	2430	2920	2	490	990	1480	1980	2470	2970	2	500	990	1490	1990	2490	2980	2	360	720	1080	1440	1800	2160	2	360	720	1080	1440	1800	2160
3	40	560	1120	1690	2250	2810	3370	3	570	1150	1720	2290	2870	3440	3	580	1150	1730	2310	2880	3460	3	420	830	1250	1670	2080	2400	3	420	830	1250	1670	2080	2400
4	45	640	1270	1910	2550	3180	3400	4	650	1300	1950	2600	3250	3800	4	650	1310	1960	2620	3270	3920	4	470	940	1420	1890	2360	2400	4	470	940	1420	1890	2360	2400
5	50	710	1420	2130	2840	3400		5	730	1450	2180	2900	3630	3800	5	730	1460	2190	2920	3650	4000	5	530	1050	1580	2110	2400		5	530	1050	1580	2110	2400	
6	55	780	1560	2340	3130	3400		6	800	1600	2400	3200	3800		6	810	1610	2420	3230	4000		6	580	1160	1740	2320	2400		6	580	1160	1740	2320	2400	
7	60	850	1700	2560	3400			7	870	1750	2620	3500	3800		7	880	1760	2640	3520	4000		7	630	1260	1900	2400			7	630	1260	1900	2400		
8	66	930	1870	2800	3400			8	960	1920	2880	3800			8	970	1940	2910	3880	4000		8	690	1380	2080	2400			8	690	1380	2080	2400		
9	70	990	1980	2970	3400			9	1020	2040	3050	3800			9	1030	2050	3080	4000			9	730	1460	2190	2400			9	730	1460	2190	2400		
10	80	1120	2240	3360	3400			10	1160	2320	3470	3800			10	1170	2340	3510	4000			10	830	1650	2400				10	830	1650	2400			
11	90	1250	2500	3400				11	1290	2590	3800				11	1310	2620	3920	4000			11	920	1830	2400				11	920	1830	2400			
12	100	1370	2740	3400				12	1420	2850	3800				12	1440	2880	4000				12	1000	2010	2400				12	1000	2010	2400			
13	110	1490	2980	3400				13	1550	3110	3800				13	1570	3150	4000				13	1090	2170	2400				13	1090	2170	2400			
14	120	1600	3210	3400				14	1680	3360	3800				14	1700	3400	4000				14	1170	2330	2400				14	1170	2330	2400			
15	150	1930	3400					15	2030	3800					15	2070	4000					15	1380	2400					15	1380	2400				
16	200	2400	3400					16	2560	3800					16	2620	4000					16	1690	2400					16	1690	2400				
17	250	2810	3400					17	3030	3800					17	3110	4000					17	1950	2400					17	1950	2400				
18	280	3030	3400					18	3290	3800					18	3380	4000					18	2090	2400					18	2090	2400				

RELEVANT TO VACUUM BRAKED TRAINS THE FOLLOWING RESTRICTIONS MAY NOT BE EXCEEDED:

a. Maximum train speed of 60 km/h.
b. Maximum average axle load per train of 18.5 ton per axle.
c. Maximum axle load per truck per axle of 20 ton.
d. Maximum train length of 40 trucks (160 axles).
e. Maximum train length on specific line as restricted in WTB.
f. Maximum axle load restriction on specific line.
g. Maximum total load of 2560 ton on category 1 lines where SASKOP-couplers are used.
h. Maximum number of locomotives in a consist is four (4).

Notes :

a. 'GC' Means Gradient Category.
b. Refer to point 5.9 below for Load Tables relevant to Passenger Trains.

This document is classified as 'Confidential' and as such is authorized and approved for general distribution within Transnet Freight Rail. (The document and/or any information contained herein may not be disclosed outside of TFR without prior permission first being obtained from the Principal Engineer (Train Design Technologies))

5.4 GENERAL LOAD TABLES FOR 7E/10E LOCOMOTIVES

LOAD TABLES FOR 7E/10E LOCOMOTIVES
AIRBRAKE - 50 WAGON TRAIN LENGTH MAXIMUM
VACUUM BRAKE - 40 WAGON TRAIN LENGTH MAXIMUM

TONNAGE RATINGS BY LINE PER LOCOMOTIVE

7E/10E ELECTRIC LOCOMOTIVES

VACUUM BRAKED TRAINS ≤ 10 TONS PER AXLE								VACUUM BRAKED TRAINS > 10 TONS PER AXLE							AIRBRAKE TRAINS >2 to 7 t/a							AIRBRAKE TRAINS >7 to 12.5 t/a						
CLASS : 7E/10E								CLASS : 7E/10E							CLASS : 7E/10E							CLASS : 7E/10E						
		(NUMBER)							(NUMBER)							(NUMBER)							(NUMBER)					
GC	GRAD.	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6
1	30	690	1380	1600	-	-	-	1	730	1460	2190	2910	-	-	1	610	1220	1400	-	-	-	1	710	1420	2130	2500	-	-
2	35	810	1600					2	860	1720	2580	2960			2	700	1400					2	840	1670	2500	2500		
3	40	920	1600					3	990	1980	2960				3	790	1400					3	960	1920	2500			
4	45	1030	1600					4	1110	2230	2960				4	880	1400					4	1080	2150	2500			
5	50	1140	1600					5	1240	2470	2960				5	950	1400					5	1190	2380	2500			
6	55	1240	1600					6	1360	2710	2960				6	1030	1400					6	1300	2500				
7	60	1340	1600					7	1470	2950	2960				7	1100	1400					7	1410	2500				
8	66	1460	1600					8	1610	2960					8	1180	1400					8	1540	2500				
9	70	1530	1600					9	1700	2960					9	1230	1400					9	1620	2500				
10	80	1600						10	1920	2960					10	1340	1400					10	1820	2500				
11	90	1600						11	2130	2960					11	1400						11	2010	2500				
12	100	1600						12	2330	2960					12	1400						12	2190	2500				
13	110	1600						13	2520	2960					13	1400						13	2360	2500				
14	120	1600						14	2710	2960					14	1400						14	2500					
15	150	1600						15	2960						15	1400						15	2500					
16	200	1600						16	2960						16	1400						16	2500					
17	250	1600						17	2960						17	1400						17	2500					
18	280	1600						18	2960						18	1400						18	2500					

7E/10E ELECTRIC LOCOMOTIVES

AIRBRAKE TRAINS >12.5 to 17 t/a								AIRBRAKE TRAINS >17 to 19 t/a							AIRBRAKE TRAINS >19 to 20 t/a							MOTOR CAR TRAINS, 8 to 12 t/a #						
CLASS : 7E/10E								CLASS : 7E/10E							CLASS : 7E/10E							CLASS : 7E/10E						
		(Number)							(Number)							(Number)							(Number)					
GC	GRAD.	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6
1	30	740	1480	2210	2950	-	-	1	750	1500	2250	2990	-	-	1	750	1500	2260	3010	-	-	1	550	1100	1650	2200	-	-
2	35	870	1740	2620	3400			2	890	1770	2660	3540			2	890	1780	2670	3560			2	650	1300	1950	2400		
3	40	1000	2010	3010	3400			3	1020	2040	3070	3800			3	1030	2060	3080	4000			3	750	1500	2250	2400		
4	45	1130	2270	3400				4	1160	2310	3470	3800			4	1160	2330	3490	4000			4	850	1690	2400			
5	50	1260	2520	3780				5	1290	2570	3800				5	1300	2590	3890	4000			5	940	1880	2400			
6	55	1380	2770	4150				6	1420	2830	3800				6	1430	2850	4000				6	1030	2070	2400			
7	60	1510	3010	4520				7	1540	3090	3800				7	1560	3110	4000				7	1120	2250	2400			
8	66	1650	3300	4950				8	1690	3390	3800				8	1710	3420	4000				8	1230	2400				
9	70	1740	3400					9	1790	3590	3800				9	1810	3620	4000				9	1300	2400				
10	80	1970	3400					10	2040	3800					10	2060	4000					10	1460	2400				
11	90	2190	3400					11	2270	3800					11	2300	4000					11	1620	2400				
12	100	2410	3400					12	2500	3800					12	2530	4000					12	1770	2400				
13	110	2610	3400					13	2720	3800					13	2760	4000					13	1910	2400				
14	120	2810	3400					14	2940	3800					14	2980	4000					14	2050	2400				
15	150	3370	3400					15	3550	3800					15	3610	4000					15	2400					
16	200	3400						16	3800						16	4000						16	2400					
17	250	3400						17	3800						17	4000						17	2400					
18	280	3400						18	3800						18	4000						18	2400					

RELEVANT TO VACUUM BRAKED TRAINS THE FOLLOWING RESTRICTIONS MAY NOT BE EXCEEDED:

a. Maximum train speed of 60 km/h.
b. Maximum average axle load per train of 18.5 ton per axle.
c. Maximum axle load per truck per axle of 20 ton.
d. Maximum train length of 40 trucks (160 axles).
e. Maximum train length on specific line as restricted in WTB.
f. Maximum axle load restriction on specific line.
g. Maximum total load of 2560 ton on category 1 lines where SASKOP-couplers are used.
h. Maximum number of locomotives in a consist is four (4).

Notes :

a. 'GC' Means Gradient Category.
b. Refer to point 5.9 below for Load Tables relevant to Passenger Trains.

This document is classified as 'Confidential' and as such is authorized and approved for general distribution within Transnet Freight Rail. (The document and/or any information contained herein may not be disclosed outside of TFR without prior permission first being obtained from the Principal Engineer (Train Design Technologies))

5.5 GENERAL LOAD TABLES FOR 8E LOCOMOTIVES

LOAD TABLES FOR 8E LOCOMOTIVES
AIRBRAKE - 50 WAGON TRAIN LENGTH MAXIMUM
VACUUM BRAKE - 40 WAGON TRAIN LENGTH MAXIMUM

TONNAGE RATINGS BY LINE PER LOCOMOTIVE

8E ELECTRIC LOCOMOTIVES

VACUUM BRAKED TRAINS ≤ 10 TONS PER AXLE								VACUUM BRAKED TRAINS > 10 TONS PER AXLE							AIRBRAKE TRAINS >2 to 7 t/a							AIRBRAKE TRAINS >7 to 12.5 t/a						
CLASS : 8E								CLASS : 8E							CLASS : 8E							CLASS : 8E						
		(NUMBER)							(NUMBER)							(NUMBER)							(NUMBER)					
GC	GRAD.	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6
1	30	140	290	430	570	-	-	1	150	310	460	620	-	-	1	120	240	360	490	-	-	1	150	300	450	600	-	-
2	35	180	350	530	710			2	190	380	570	760			2	150	300	440	590			2	180	370	550	740		
3	40	210	420	620	830			3	230	450	680	900			3	170	340	520	690			3	220	440	650	870		
4	45	240	480	720	960			4	260	520	780	1040			4	200	390	590	780			4	250	500	750	1000		
5	50	270	540	800	1070			5	290	590	880	1180			5	220	430	650	870			5	280	560	850	1130		
6	55	300	590	890	1190			6	330	660	980	1310			6	240	480	710	950			6	310	630	940	1250		
7	60	320	650	970	1300			7	360	720	1080	1440			7	260	510	770	1030			7	340	690	1030	1370		
8	66	360	710	1070	1430			8	400	800	1200	1590			8	280	560	840	1120			8	380	760	1140	1510		
9	70	380	750	1130	1510			9	420	850	1270	1690			9	290	590	880	1170			9	400	800	1200	1610		
10	80	430	850	1280	1600			10	480	970	1450	1930			10	320	650	970	1300			10	460	910	1370	1820		
11	90	470	940	1420	1600			11	540	1080	1620	2170			11	350	710	1060	1400			11	510	1020	1520	2030		
12	100	520	1030	1550	1600			12	600	1190	1790	2390			12	380	760	1140	1400			12	560	1120	1670	2230		
13	110	560	1110	1600				13	650	1300	1950	2600			13	400	810	1210	1400			13	610	1210	1820	2420		
14	120	600	1190	1600				14	700	1400	2110	2810			14	430	850	1280	1400			14	650	1300	1950	2500		
15	150	700	1400	1600				15	840	1690	2530	2960			15	480	970	1400				15	770	1550	2320	2500		
16	200	840	1600					16	1050	2100	2960				16	550	1110	1400				16	950	1890	2500			
17	250	960	1600					17	1230	2450	2960				17	610	1210	1400				17	1090	2180	2500			
18	280	1010	1600					18	1320	2640	2960				18	630	1260	1400				18	1160	2330	2500			

8E ELECTRIC LOCOMOTIVES

AIRBRAKE TRAINS >12.5 to 17 t/a								AIRBRAKE TRAINS >17 to 19 t/a							AIRBRAKE TRAINS >19 to 20 t/a							MOTOR CAR TRAINS, 8 to 12 t/a #						
CLASS : 8E								CLASS : 8E							CLASS : 8E							CLASS : 8E						
		(Number)							(Number)							(Number)							(Number)					
GC	GRAD.	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6
1	30	160	310	470	630	-	-	1	160	320	480	640	-	-	1	160	320	480	640	-	-	1	100	210	310	420	-	-
2	35	190	390	580	780			2	200	400	590	790			2	200	400	600	800			2	130	270	400	530		
3	40	230	460	690	920			3	240	470	710	940			3	240	470	710	950			3	160	320	480	640		
4	45	270	530	800	1060			4	270	550	820	1090			4	270	550	820	1100			4	190	370	560	750		
5	50	300	600	900	1210			5	310	620	930	1240			5	310	620	930	1250			5	210	430	640	850		
6	55	340	670	1010	1340			6	340	690	1030	1380			6	350	700	1040	1390			6	240	480	720	960		
7	60	370	740	1110	1480			7	380	760	1140	1520			7	380	770	1150	1530			7	260	530	790	1050		
8	66	410	820	1230	1640			8	420	840	1270	1690			8	430	850	1280	1700			8	290	590	880	1170		
9	70	440	870	1310	1740			9	450	900	1350	1800			9	450	910	1360	1810			9	310	620	930	1250		
10	80	500	1000	1500	1990			10	520	1030	1550	2060			10	520	1040	1570	2090			10	360	710	1070	1430		
11	90	560	1120	1680	2240			11	580	1160	1740	2320			11	590	1180	1760	2350			11	400	800	1200	1600		
12	100	620	1240	1860	2470			12	640	1290	1930	2580			12	650	1310	1960	2610			12	440	880	1330	1770		
13	110	680	1350	2030	2700			13	710	1410	2120	2820			13	720	1430	2150	2860			13	480	960	1450	1930		
14	120	730	1460	2190	2920			14	770	1530	2300	3060			14	780	1550	2330	3110			14	520	1040	1560	2080		
15	150	890	1770	2660	3400			15	940	1870	2810	3740			15	950	1900	2860	3810			15	620	1250	1870	2400		
16	200	1110	2230	3340	3400			16	1190	2380	3570	3800			16	1220	2430	3650	4000			16	770	1550	2320	2400		
17	250	1310	2620	3400				17	1420	2830	3800				17	1450	2900	4000				17	900	1790	2400			
18	280	1410	2830	3400				18	1540	3080	3800				18	1580	3160	4000				18	960	1920	2400			

RELEVANT TO VACUUM BRAKED TRAINS THE FOLLOWING RESTRICTIONS MAY NOT BE EXCEEDED:

a. Maximum train speed of 60 km/h.

b. Maximum average axle load per train of 18.5 ton per axle.

c. Maximum axle load per truck per axle of 20 ton.

d. Maximum train length of 40 trucks (160 axles).

e. Maximum train length on specific line as restricted in WTB.
f. Maximum axle load restriction on specific line.
g. Maximum total load of 2560 ton on category 1 lines where SASKOP-couplers are used.
h. Maximum number of locomotives in a consist is four (4).

Notes :
a. 'GC' Means Gradient Category.
b. Refer to point 5.9 below for Load Tables relevant to Passenger Trains.
b. A table depicting "Downgrade Speed Restrictions" relevant to the various train types has been included for trains hauled by 8E electric locomotives.
(In essence this table is a reflection of G.A 1019.8 and 1019.10 (i.e. Downgrade speed restrictions for locomotives with insufficient electric brake). Note that the 8E does have any electrical braking facility.)
c. The downgrade speeds reflected for the 8E locomotives in this document will remain valid, unless overruled and/or more restrictive speed limts have been specified for whatsoever reason.
d. Trains hauled by 8E locomotives are not authorised on downgrades steeper than 1 in 50. Extremely restrictive downgrade speed restrictions would need to be imposed to prevent heat input into the wheels from exceeding 12KW.

8E ELECTRICAL LOCOMOTIVES DOWNGRADE SPEED RESTRICTIONS

GRADIENT CATEGORY	RULING DOWN GRADIENT	VAC ≤ 10T/A	VAC > 10T/A	AIR > 2 - 7 T/A	AIR > 7 - 12.5 T/A	AIR > 12.5 to 17 T/A	AIR > 17- 19 T/A	AIR > 19 - 20 T/A	AIR MOTOR CAR
1	30 - 34	NOT AUTHORISED							
2	35 - 39								
3	40 - 44								
4	45 - 49								
5	50 - 54	20	20	20	20	20	20	20	20
6	55 - 59	20	20	20	20	20	20	20	20
7	60 - 65	25	25	25	25	25	25	25	25
8	66 - 70	25	25	25	25	25	25	25	25
9	70 - 79	25	25	25	25	25	25	25	25
10	80 - 89	35	35	35	35	35	35	35	35
11	90 - 99	35	35	35	35	35	35	35	35
12	100 - 109	50	50	50	50	50	45	45	50
13	110 - 119	50	50	50	50	50	50	50	50
14	120 - 149	50	50	50	50	50	50	50	50
15	150 - 179	50	50	50	50	50	50	50	50
16	180 - 249	50	50	50	50	50	50	50	50
17	250 - 279	50	50	50	50	50	50	50	50
18	280 →	50	50	50	50	50	50	50	50

This document is classified as 'Confidential' and as such is authorized and approved for general distribution within Transnet Freight Rail. (The document and/or any information contained herein may not be disclosed outside of TFR without prior permission first being obtained from the Principal Engineer (Train Design Technologies))

5.6 GENERAL LOAD TABLES FOR 18E LOCOMOTIVES

LOAD TABLES FOR 18E LOCOMOTIVES
AIRBRAKE - 50 WAGON TRAIN LENGTH MAXIMUM
VACUUM BRAKE - 40 WAGON TRAIN LENGTH MAXIMUM

TONNAGE RATINGS BY LINE PER LOCOMOTIVE

18E ELECTRIC LOCOMOTIVES

VACUUM BRAKED TRAINS ≤ 10 TONS PER AXLE								VACUUM BRAKED TRAINS > 10 TONS PER AXLE							AIRBRAKE TRAINS >2 to 7 t/a							AIRBRAKE TRAINS >7 to 12.5 t/a						
CLASS : 18E								CLASS : 18E							CLASS : 18E							CLASS : 18E						
		(NUMBER)							(NUMBER)							(NUMBER)							(NUMBER)					
GC	GRAD.	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6
1	30	400	830	1280	1600	-	-	1	420	880	1360	1840	-	-	1	350	730	1130	1400	-	-	1	410	850	1320	1790	2260	2500
2	35	470	970	1510	1600			2	500	1040	1610	2170			2	410	850	1310	1400			2	490	1010	1560	2110	2500	
3	40	540	1120	1600				3	580	1200	1850	2500			3	460	960	1400				3	560	1160	1790	2430	2500	
4	45	610	1250	1600				4	650	1350	2090	2820			4	510	1060	1400				4	630	1300	2020	2500		
5	50	670	1380	1600				5	730	1500	2320	2960			5	560	1160	1400				5	700	1450	2240	2500		
6	55	730	1510	1600				6	800	1650	2550	2960			6	600	1250	1400				6	770	1580	2450	2500		
7	60	790	1600					7	870	1790	2770	2960			7	650	1330	1400				7	830	1720	2500			
8	66	860	1600					8	950	1960	2960				8	690	1400					8	910	1880	2500			
9	70	910	1600					9	1010	2070	2960				9	720	1400					9	960	1980	2500			
10	80	1010	1600					10	1140	2340	2960				10	790	1400					10	1080	2220	2500			
11	90	1110	1600					11	1260	2600	2960				11	860	1400					11	1190	2450	2500			
12	100	1210	1600					12	1380	2850	2960				12	910	1400					12	1300	2500				
13	110	1300	1600					13	1500	2960					13	970	1400					13	1400	2500				
14	120	1380	1600					14	1610	2960					14	1010	1400					14	1500	2500				
15	150	1600						15	1920	2960					15	1140	1400					15	1770	2500				
16	200	1600						16	2370	2960					16	1290	1400					16	2140	2500				
17	250	1600						17	2750	2960					17	1400						17	2450	2500				
18	280	1600						18	2950						18	1400						18	2500					

18E ELECTRIC LOCOMOTIVES

AIRBRAKE TRAINS >12.5 to 17 t/a								AIRBRAKE TRAINS >17 to 19 t/a							AIRBRAKE TRAINS >19 to 20 t/a							MOTOR CAR TRAINS, 8 to 12 t/a #						
CLASS : 18E								CLASS : 18E							CLASS : 18E							CLASS : 18E						
		(Number)							(Number)							(Number)							(Number)					
GC	GRAD.	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6
1	30	430	890	1370	1860	2350	2840	1	440	900	1390	1890	2380	2880	1	440	900	1400	1900	2390	2890	1	320	650	1020	1380	1740	2100
2	35	510	1050	1630	2210	2780	3360	2	520	1070	1660	2240	2830	3410	2	520	1080	1660	2250	2840	3430	2	380	780	1210	1640	2070	2400
3	40	590	1220	1880	2540	3210	3400	3	600	1240	1910	2590	3270	3800	3	600	1240	1930	2610	3290	3970	3	440	900	1400	1890	2390	2400
4	45	670	1370	2120	2870	3400		4	680	1400	2170	2930	3700	3800	4	680	1410	2180	2950	3720	4000	4	500	1020	1580	2140	2400	
5	50	740	1530	2370	3200	3400		5	760	1560	2420	3270	3800		5	760	1580	2430	3290	4000		5	550	1140	1760	2400		
6	55	820	1680	2600	3400			6	840	1720	2660	3600	3800		6	840	1740	2680	3630	4000		6	610	1250	1940	2400		
7	60	890	1830	2830	3400			7	910	1880	2910	3800			7	920	1900	2930	3960	4000		7	660	1360	2110	2400		
8	66	980	2010	3110	3400			8	1000	2070	3190	3800			8	1010	2090	3220	4000			8	720	1490	2310	2400		
9	70	1030	2130	3280	3400			9	1060	2190	3380	3800			9	1070	2210	3410	4000			9	760	1580	2400			
10	80	1170	2410	3400				10	1210	2490	3800				10	1220	2510	3880	4000			10	860	1780	2400			
11	90	1300	2680	3400				11	1350	2780	3800				11	1360	2810	4000				11	960	1970	2400			
12	100	1430	2940	3400				12	1490	3060	3800				12	1500	3100	4000				12	1050	2160	2400			
13	110	1550	3200	3400				13	1620	3330	3800				13	1640	3380	4000				13	1130	2340	2400			
14	120	1670	3400					14	1750	3800					14	1770	3650	4000				14	1220	2400				
15	150	2010	3400					15	2120	3800					15	2150	4000					15	1440	2400				
16	200	2500	3400					16	2670	3800					16	2730	4000					16	1770	2400				
17	250	2930	3400					17	3160	3800					17	3240	4000					17	2030	2400				
18	280	3160	3400					18	3420	3800					18	3520	4000					18	2170	2400				

RELEVANT TO VACUUM BRAKED TRAINS THE FOLLOWING RESTRICTIONS MAY NOT BE EXCEEDED:

a. Maximum train speed of 60 km/h.
b. Maximum average axle load per train of 18.5 ton per axle.
c. Maximum axle load per truck per axle of 20 ton.
d. Maximum train length of 40 trucks (160 axles).
e. Maximum train length on specific line as restricted in WTB.
f. Maximum axle load restriction on specific line.
g. Maximum total load of 2560 ton on category 1 lines where SASKOP-couplers are used.
h. Maximum number of locomotives in a consist is four (4).

Notes :

a. 'GC' Means Gradient Category.
b. Refer to point 5.9 below for 18E Electric Locomotive Load Tables relevant to passenger trains.

This document is classified as 'Confidential' and as such is authorized and approved for general distribution within Transnet Freight Rail. (The document and/or any information contained herein may not be disclosed outside of TFR without prior permission first being obtained from the Principal Engineer (Train Design Technologies))

5.7 GENERAL LOAD TABLES FOR 20E LOCOMOTIVES

LOAD TABLES FOR 20E LOCOMOTIVES
AIRBRAKE - 50 WAGON TRAIN LENGTH MAXIMUM
VACUUM BRAKE - 40 WAGON TRAIN LENGTH MAXIMUM

TONNAGE RATINGS BY LINE PER LOCOMOTIVE

20E ELECTRIC LOCOMOTIVES

VACUUM BRAKED TRAINS ≤ 10 TONS PER AXLE								VACUUM BRAKED TRAINS > 10 TONS PER AXLE							AIRBRAKE TRAINS >2 to 7 t/a							AIRBRAKE TRAINS >7 to 12.5 t/a						
CLASS : 20E								CLASS : 20E							CLASS : 20E							CLASS : 20E						
		(NUMBER)							(NUMBER)							(NUMBER)							(NUMBER)					
GC	GRAD.	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6
1	30	-	-	-	-	-	-	1	-	-	-	-	-	-	1	600	1200	1400	-	-	-	1	690	1390	2080	2500	-	-
2	35							2							2	690	1370	1400				2	810	1620	2430	2500		
3	40							3							3	770	1400					3	920	1850	2500			
4	45							4							4	850	1400					4	1030	2070	2500			
5	50							5							5	920	1400					5	1140	2280	2500			
6	55							6							6	990	1400					6	1250	2490	2500			
7	60							7							7	1060	1400					7	1350	2500				
8	66							8							8	1130	1400					8	1470	2500				
9	70							9							9	1180	1400					9	1540	2500				
10	80							10							10	1280	1400					10	1730	2500				
11	90							11							11	1380	1400					11	1900	2500				
12	100							12							12	1400						12	2070	2500				
13	110							13							13	1400						13	2230	2500				
14	120							14							14	1400						14	2380	2500				
15	150							15							15	1400						15	2500					
16	200							16							16	1400						16	2500					
17	250							17							17	1400						17	2500					
18	280							18							18	1400						18	2500					

20E ELECTRIC LOCOMOTIVES

AIRBRAKE TRAINS >12.5 to 17 t/a								AIRBRAKE TRAINS >17 to 19 t/a							AIRBRAKE TRAINS >19 to 20 t/a							MOTOR CAR TRAINS, 8 to 12 t/a #						
CLASS : 20E								CLASS : 20E							CLASS : 20E							CLASS : 20E						
		(Number)							(Number)							(Number)							(Number)					
GC	GRAD.	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6
1	30	720	1440	2150	2870	3400	-	1	730	1460	2180	2910	3640	3800	1	730	1460	2190	2920	3660	4000	1	540	1080	1630	2170	2400	-
2	35	840	1690	2530	3370	3400		2	860	1710	2570	3430	3800		2	860	1720	2580	3440	4000		2	640	1270	1910	2400		
3	40	970	1930	2900	3400			3	980	1970	2950	3800			3	990	1980	2970	3950	4000		3	730	1460	2190	2400		
4	45	1090	2170	3260	3400			4	1110	2220	3320	3800			4	1110	2230	3340	4000			4	820	1640	2400			
5	50	1200	2410	3400				5	1230	2460	3690	3800			5	1240	2480	3720	4000			5	910	1820	2400			
6	55	1320	2640	3400				6	1350	2700	3800				6	1360	2720	4000				6	990	1990	2400			
7	60	1430	2870	3400				7	1470	2940	3800				7	1480	2960	4000				7	1080	2160	2400			
8	66	1570	3140	3400				8	1610	3220	3800				8	1620	3250	4000				8	1180	2350	2400			
9	70	1660	3310	3400				9	1700	3410	3800				9	1720	3440	4000				9	1240	2400				
10	80	1870	3400					10	1930	3800					10	1950	3900	4000				10	1390	2400				
11	90	2080	3400					11	2150	3800					11	2170	4000					11	1540	2400				
12	100	2270	3400					12	2360	3800					12	2390	4000					12	1680	2400				
13	110	2470	3400					13	2570	3800					13	2600	4000					13	1810	2400				
14	120	2650	3400					14	2770	3800					14	2810	4000					14	1940	2400				
15	150	3170	3400					15	3340	3800					15	3400	4000					15	2300	2400				
16	200	3400						16	3800						16	4000						16	2400					
17	250	3400						17	3800						17	4000						17	2400					
18	280	3400						18	3800						18	4000						18	2400					

RELEVANT TO VACUUM BRAKED TRAINS THE FOLLOWING RESTRICTIONS MAY NOT BE EXCEEDED:

a. Maximum train speed of 60 km/h.
b. Maximum average axle load per train of 18.5 ton per axle.
c. Maximum axle load per truck per axle of 20 ton.
d. Maximum train length of 40 trucks (160 axles).
e. Maximum train length on specific line as restricted in WTB.
f. Maximum axle load restriction on specific line.
g. Maximum total load of 2560 ton on category 1 lines where SASKOP-couplers are used.
h. Maximum number of locomotives in a consist is four (4).

Notes :

a. 'GC' Means Gradient Category.
b. Refer to point 5.9 below for 18E Electric Locomotive Load Tables relevant to passenger trains.

This document is classified as 'Confidential' and as such is authorized and approved for general distribution within Transnet Freight Rail. (The document and/or any information contained herein may not be disclosed outside of TFR without prior permission first being obtained from the Principal Engineer (Train Design Technologies))

5.8 GENERAL LOAD TABLES FOR 22E LOCOMOTIVES

LOAD TABLES FOR 22E LOCOMOTIVES
AIRBRAKE - 50 WAGON TRAIN LENGTH MAXIMUM
VACUUM BRAKE - 40 WAGON TRAIN LENGTH MAXIMUM

TONNAGE RATINGS BY LINE PER LOCOMOTIVE

22E ELECTRIC LOCOMOTIVES

VACUUM BRAKED TRAINS ≤ 10 TONS PER AXLE								VACUUM BRAKED TRAINS > 10 TONS PER AXLE							AIRBRAKE TRAINS >2 to 7 t/a							AIRBRAKE TRAINS >7 to 12.5 t/a						
CLASS : 22E								CLASS : 22E							CLASS : 22E							CLASS : 22E						
		(NUMBER)							(NUMBER)							(NUMBER)							(NUMBER)					
GC	GRAD.	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6
1	30	-	-	-	-	-	-	1	-	-	-	-	-	-	1	890	1400	-	-	-	-	1	1030	2060	2500	-	-	-
2	35							2							2	1020	1400					2	1210	2410	2500			
3	40							3							3	1140	1400					3	1370	2500				
4	45							4							4	1260	1400					4	1540	2500				
5	50							5							5	1370	1400					5	1700	2500				
6	55							6							6	1400						6	1850	2500				
7	60							7							7	1400						7	2010	2500				
8	66							8							8	1400						8	2180	2500				
9	70							9							9	1400						9	2300	2500				
10	80							10							10	1400						10	2500					
11	90							11							11	1400						11	2500					
12	100							12							12	1400						12	2500					
13	110							13							13	1400						13	2500					
14	120							14							14	1400						14	2500					
15	150							15							15	1400						15	2500					
16	200							16							16	1400						16	2500					
17	250							17							17	1400						17	2500					
18	280							18							18	1400						18	2500					

22E ELECTRIC LOCOMOTIVES

AIRBRAKE TRAINS >12.5 to 17 t/a								AIRBRAKE TRAINS >17 to 19 t/a							AIRBRAKE TRAINS >19 to 20 t/a							MOTOR CAR TRAINS, 8 to 12 t/a #						
CLASS : 22E								CLASS : 22E							CLASS : 22E							CLASS : 22E						
		(Number)							(Number)							(Number)							(Number)					
GC	GRAD.	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6
1	30	1070	2140	3210	3400	-	-	1	1080	2170	3250	3800	-	-	1	1090	2180	3260	4000	-	-	1	810	1610	2420	2400	-	-
2	35	1260	2510	3400				2	1270	2550	3800				2	1280	2560	3840	4000			2	950	1900	2400			
3	40	1440	2880	3400				3	1460	2930	3800				3	1470	2940	4000				3	1090	2170	2400			
4	45	1620	3230	3400				4	1650	3300	3800				4	1660	3320	4000				4	1220	2400				
5	50	1790	3400					5	1830	3660	3800				5	1840	3690	4000				5	1350	2400				
6	55	1970	3400					6	2010	3800					6	2030	4000					6	1480	2400				
7	60	2140	3400					7	2190	3800					7	2210	4000					7	1600	2400				
8	66	2340	3400					8	2400	3800					8	2420	4000					8	1750	2400				
9	70	2470	3400					9	2540	3800					9	2560	4000					9	1850	2400				
10	80	2780	3400					10	2870	3800					10	2900	4000					10	2070	2400				
11	90	3090	3400					11	3200	3800					11	3230	4000					11	2290	2400				
12	100	3390	3400					12	3520	3800					12	3560	4000					12	2400					
13	110	3400						13	3800						13	3880	4000					13	2400					
14	120	3400						14	3800						14	4000						14	2400					
15	150	3400						15	3800						15	4000						15	2400					
16	200	3400						16	3800						16	4000						16	2400					
17	250	3400						17	3800						17	4000						17	2400					
18	280	3400						18	3800						18	4000						18	2400					

RELEVANT TO VACUUM BRAKED TRAINS THE FOLLOWING RESTRICTIONS MAY NOT BE EXCEEDED:

a. Maximum train speed of 60 km/h.
b. Maximum average axle load per train of 18.5 ton per axle.
c. Maximum axle load per truck per axle of 20 ton.
d. Maximum train length of 40 trucks (160 axles).
e. Maximum train length on specific line as restricted in WTB.
f. Maximum axle load restriction on specific line.
g. Maximum total load of 2560 ton on category 1 lines where SASKOP-couplers are used.
h. Maximum number of locomotives in a consist is four (4).

Notes : a. 'GC' Means Gradient Category.

This document is classified as 'Confidential' and as such is authorized and approved for general distribution within Transnet Freight Rail. (The document and/or any information contained herein may not be disclosed outside of TFR without prior permission first being obtained from the Principal Engineer (Train Design Technologies))

5.9 GENERAL LOAD TABLES FOR 33D LOCOMOTIVES

LOAD TABLES FOR 33 CLASS DIESEL LOCOMOTIVES
AIRBRAKE - 50 WAGON TRAIN LENGTH MAXIMUM
VACUUM BRAKE - 40 WAGON TRAIN LENGTH MAXIMUM

TONNAGE RATINGS BY LINE PER LOCOMOTIVE
33 CLASS DIESEL LOCOMOTIVES

VACUUM BRAKED TRAINS ≤ 10 TONS PER AXLE								VACUUM BRAKED TRAINS > 10 TONS PER AXLE							AIRBRAKE TRAINS >2 to 7 t/a							AIRBRAKE TRAINS >7 to 12.5 t/a						
CLASS : 33D								CLASS : 33D							CLASS : 33D							CLASS : 33D						
		(NUMBER)							(NUMBER)							(NUMBER)							(NUMBER)					
GC	GRAD.	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6
1	30	340	690	1030	1380	-	-	1	360	730	1090	1460	-	-	1	300	600	900	1200	-	-	1	360	710	1070	1420	-	-
2	35	410	810	1220	1600			2	430	870	1300	1740			2	350	700	1050	1400			2	420	840	1270	1690		
3	40	470	940	1410	1600			3	500	1010	1510	2010			3	400	800	1200	1400			3	490	970	1460	1950		
4	45	530	1050	1580	1600			4	570	1140	1710	2280			4	440	890	1330	1400			4	550	1100	1650	2200		
5	50	580	1170	1600				5	640	1270	1910	2540			5	490	970	1400				5	610	1220	1830	2450		
6	55	640	1280	1600				6	700	1400	2100	2800			6	530	1050	1400				6	670	1340	2010	2500		
7	60	690	1380	1600				7	760	1520	2290	2960			7	560	1120	1400				7	730	1460	2190	2500		
8	66	750	1510	1600				8	840	1670	2510	2960			8	600	1210	1400				8	800	1590	2390	2500		
9	70	790	1590	1600				9	880	1770	2650	2960			9	630	1260	1400				9	840	1680	2500			
10	80	890	1600					10	1000	2000	2960				10	690	1390	1400				10	950	1890	2500			
11	90	980	1600					11	1110	2220	2960				11	750	1400					11	1050	2090	2500			
12	100	1060	1600					12	1220	2440	2960				12	800	1400					12	1140	2280	2500			
13	110	1140	1600					13	1320	2640	2960				13	850	1400					13	1230	2470	2500			
14	120	1210	1600					14	1420	2840	2960				14	890	1400					14	1320	2500				
15	150	1420	1600					15	1700	2960					15	1000	1400					15	1560	2500				
16	200	1600						16	2090	2960					16	1140	1400					16	1890	2500				
17	250	1600						17	2430	2960					17	1240	1400					17	2170	2500				
18	280	1600						18	2610	2960					18	1280	1400					18	2310	2500				

33 CLASS DIESEL LOCOMOTIVES

AIRBRAKE TRAINS >12.5 to 17 t/a								AIRBRAKE TRAINS >17 to 19 t/a							AIRBRAKE TRAINS >19 to 20 t/a							MOTOR CAR TRAINS, 8 to 12 t/a #						
CLASS : 33D								CLASS : 33D							CLASS : 33D							CLASS : 33D						
		(Number)							(Number)							(Number)							(Number)					
GC	GRAD.	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6
1	30	370	740	1110	1480	-	-	1	380	750	1130	1500	-	-	1	380	750	1130	1510	-	-	1	270	540	810	1080	-	-
2	35	440	880	1320	1760			2	450	900	1350	1800			2	450	900	1350	1800			2	320	650	970	1290		
3	40	510	1020	1530	2050			3	520	1040	1560	2080			3	520	1050	1570	2100			3	380	750	1130	1510		
4	45	580	1160	1740	2320			4	590	1180	1780	2370			4	600	1190	1790	2380			4	430	860	1280	1710		
5	50	650	1300	1940	2590			5	660	1330	1990	2650			5	670	1330	2000	2670			5	480	960	1440	1910		
6	55	710	1430	2140	2860			6	730	1460	2190	2930			6	740	1470	2210	2950			6	530	1060	1580	2110		
7	60	780	1560	2340	3120			7	800	1600	2400	3200			7	810	1610	2420	3220			7	580	1150	1730	2300		
8	66	860	1710	2570	3400			8	880	1760	2640	3520			8	890	1780	2660	3550			8	630	1260	1890	2400		
9	70	910	1810	2720	3400			9	930	1870	2800	3730			9	940	1880	2820	3770			9	670	1340	2000	2400		
10	80	1030	2060	3080	3400			10	1060	2120	3190	3800			10	1070	2150	3220	4000			10	760	1510	2270	2400		
11	90	1150	2290	3400				11	1190	2370	3560	3800			11	1200	2400	3600	4000			11	840	1680	2400			
12	100	1260	2520	3400				12	1310	2620	3800				12	1330	2650	3980	4000			12	920	1840	2400			
13	110	1370	2740	3400				13	1430	2860	3800				13	1450	2890	4000				13	1000	1990	2400			
14	120	1480	2950	3400				14	1540	3090	3800				14	1560	3130	4000				14	1070	2140	2400			
15	150	1770	3400					15	1870	3740	3800				15	1900	3800	4000				15	1270	2400				
16	200	2210	3400					16	2360	3800					16	2410	4000					16	1560	2400				
17	250	2590	3400					17	2800	3800					17	2860	4000					17	1800	2400				
18	280	2790	3400					18	3030	3800					18	3110	4000					18	1920	2400				

RELEVANT TO VACUUM BRAKED TRAINS THE FOLLOWING RESTRICTIONS MAY NOT BE EXCEEDED:

a. Maximum train speed of 60 km/h.
b. Maximum average axle load per train of 18.5 ton per axle.
c. Maximum axle load per truck per axle of 20 ton.
d. Maximum train length of 40 trucks (160 axles).
e. Maximum train length on specific line as restricted in WTB.
f. Maximum axle load restriction on specific line.
g. Maximum total load of 2560 ton on category 1 lines where SASKOP-couplers are used.
h. Maximum number of locomotives in a consist is four (4).

Notes :

a. 'GC' Means Gradient Category.
b. Refer to point 5.9 below for Load Tables relevant to Passenger Trains.

This document is classified as 'Confidential' and as such is authorized and approved for general distribution within Transnet Freight Rail. (The document and/or any information contained herein may not be disclosed outside of TFR without prior permission first being obtained from the Principal Engineer (Train Design Technologies))

5.10 GENERAL LOAD TABLES FOR 34D LOCOMOTIVES

LOAD TABLES FOR 34 CLASS DIESEL LOCOMOTIVES
AIRBRAKE - 50 WAGON TRAIN LENGTH MAXIMUM
VACUUM BRAKE - 40 WAGON TRAIN LENGTH MAXIMUM

TONNAGE RATINGS BY LINE PER LOCOMOTIVE

34 CLASS DIESEL LOCOMOTIVES

VACUUM BRAKED TRAINS ≤ 10 TONS PER AXLE								VACUUM BRAKED TRAINS > 10 TONS PER AXLE							AIRBRAKE TRAINS >2 to 7 t/a							AIRBRAKE TRAINS >7 to 12.5 t/a						
CLASS : 34D								CLASS : 34D							CLASS : 34D							CLASS : 34D						
		(NUMBER)							(NUMBER)							(NUMBER)							(NUMBER)					
GC	GRAD.	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6
1	30	450	900	1360	1600	-	-	1	480	960	1440	1910	-	-	1	400	790	1190	1400	-	-	1	470	930	1400	1870	2330	2500
2	35	530	1070	1600				2	570	1140	1710	2280			2	460	920	1380	1400			2	550	1110	1660	2210	2500	
3	40	610	1230	1600				3	660	1320	1970	2630			3	520	1050	1400				3	640	1270	1910	2500		
4	45	690	1380	1600				4	740	1490	2230	2960			4	580	1160	1400				4	720	1440	2160	2500		
5	50	760	1530	1600				5	830	1660	2490	2960			5	630	1270	1400				5	800	1600	2390	2500		
6	55	830	1600					6	910	1820	2730	2960			6	690	1370	1400				6	870	1750	2500			
7	60	900	1600					7	990	1980	2960				7	730	1400					7	950	1900	2500			
8	66	980	1600					8	1090	2170	2960				8	790	1400					8	1040	2080	2500			
9	70	1030	1600					9	1150	2300	2960				9	820	1400					9	1090	2190	2500			
10	80	1160	1600					10	1300	2600	2960				10	900	1400					10	1230	2460	2500			
11	90	1270	1600					11	1440	2890	2960				11	980	1400					11	1360	2500				
12	100	1380	1600					12	1580	2960					12	1040	1400					12	1480	2500				
13	110	1480	1600					13	1720	2960					13	1100	1400					13	1600	2500				
14	120	1580	1600					14	1840	2960					14	1160	1400					14	1710	2500				
15	150	1600	1600					15	2200	2960					15	1300	1400					15	2020	2500				
16	200	1600	1600					16	2720	2960					16	1400	1400					16	2500					
17	250	1600	1600					17	2960	2960					17	1400	1400					17	2500					
18	280	1600	1600					18	2960	2960					18	1400	1400					18	2500					

34 CLASS DIESEL LOCOMOTIVES

AIRBRAKE TRAINS >12.5 to 17 t/a								AIRBRAKE TRAINS >17 to 19 t/a							AIRBRAKE TRAINS >19 to 20 t/a							MOTOR CAR TRAINS, 8 to 12 t/a #						
CLASS : 34D								CLASS : 34D							CLASS : 34D							CLASS : 34D						
		(Number)							(Number)							(Number)							(Number)					
GC	GRAD.	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6
1	30	490	970	1460	1940	2430	2910	1	490	980	1480	1970	2460	2950	1	490	990	1480	1980	2470	2970	1	350	710	1060	1420	1770	2130
2	35	580	1160	1730	2310	2890	3400	2	590	1170	1760	2350	2940	3520	2	590	1180	1770	2360	2950	3540	2	430	850	1280	1700	2130	2400
3	40	670	1340	2010	2670	3340	3400	3	680	1360	2040	2720	3400	3800	3	680	1370	2050	2740	3420	4000	3	490	990	1480	1980	2400	
4	45	760	1510	2270	3030	3400		4	770	1550	2320	3090	3800		4	780	1560	2330	3110	3890	4000	4	560	1120	1680	2240	2400	
5	50	840	1690	2530	3380	3400		5	860	1730	2590	3460	3800		5	870	1740	2610	3480	4000		5	630	1250	1880	2400		
6	55	930	1860	2790	3400			6	950	1910	2860	3800			6	960	1920	2880	3840	4000		6	690	1380	2070	2400		
7	60	1010	2030	3040	3400			7	1040	2080	3120	3800			7	1050	2100	3150	4000			7	750	1500	2250	2400		
8	66	1110	2230	3340	3400			8	1150	2290	3440	3800			8	1160	2310	3470	4000			8	820	1650	2400			
9	70	1180	2360	3400				9	1210	2430	3640	3800			9	1220	2450	3670	4000			9	870	1740	2400			
10	80	1340	2670	3400				10	1380	2760	3800				10	1390	2790	4000				10	980	1970	2400			
11	90	1490	2980	3400				11	1540	3090	3800				11	1560	3120	4000				11	1090	2190	2400			
12	100	1640	3270	3400				12	1700	3400	3800				12	1720	3440	4000				12	1200	2390	2400			
13	110	1780	3400					13	1850	3710	3800				13	1880	3760	4000				13	1300	2400				
14	120	1920	3400					14	2000	3800					14	2030	4000					14	1390	2400				
15	150	2300	3400					15	2430	3800					15	2470	4000					15	1650	2400				
16	200	2870	3400					16	3060	3800					16	3130	4000					16	2020	2400				
17	250	3360	3400					17	3620	3800					17	3710	4000					17	2330	2400				
18	280	3400						18	3800						18	4000						18	2400					

RELEVANT TO VACUUM BRAKED TRAINS THE FOLLOWING RESTRICTIONS MAY NOT BE EXCEEDED:

a. Maximum train speed of 60 km/h.
b. Maximum average axle load per train of 18.5 ton per axle.
c. Maximum axle load per truck per axle of 20 ton.
d. Maximum train length of 40 trucks (160 axles).
e. Maximum train length on specific line as restricted in WTB.
f. Maximum axle load restriction on specific line.
g. Maximum total load of 2560 ton on category 1 lines where SASKOP-couplers are used.
h. Maximum number of locomotives in a consist is four (4).

Notes :

a. 'GC' Means Gradient Category.
b. Refer to point 5.9 below for Load Tables relevant to Passenger Trains.

This document is classified as 'Confidential' and as such is authorized and approved for general distribution within Transnet Freight Rail. (The document and/or any information contained herein may not be disclosed outside of TFR without prior permission first being obtained from the Principal Engineer (Train Design Technologies))

5.11 GENERAL LOAD TABLES FOR 35D LOCOMOTIVES

LOAD TABLES FOR 35 CLASS DIESEL LOCOMOTIVES
AIRBRAKE - 50 WAGON TRAIN LENGTH MAXIMUM
VACUUM BRAKE - 40 WAGON TRAIN LENGTH MAXIMUM

TONNAGE RATINGS BY LINE PER LOCOMOTIVE

35 CLASS DIESEL LOCOMOTIVES

VACUUM BRAKED TRAINS ≤ 10 TONS PER AXLE								VACUUM BRAKED TRAINS > 10 TONS PER AXLE							AIRBRAKE TRAINS >2 to 7 t/a							AIRBRAKE TRAINS >7 to 12.5 t/a						
CLASS : 35D								CLASS : 35D							CLASS : 35D							CLASS : 35D						
		(NUMBER)							(NUMBER)							(NUMBER)							(NUMBER)					
GC	GRAD.	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6
1	30	330	670	1000	1340	-	-	1	350	710	1060	1420	-	-	1	290	590	880	1170	-	-	1	340	690	1030	1380	-	-
2	35	390	790	1180	1580			2	420	840	1260	1680			2	340	680	1020	1360			2	410	820	1230	1630		
3	40	450	910	1360	1600			3	490	970	1460	1940			3	390	770	1160	1400			3	470	940	1410	1880		
4	45	510	1020	1530	1600			4	550	1100	1650	2200			4	430	860	1290	1400			4	530	1060	1590	2120		
5	50	560	1130	1600				5	610	1220	1840	2450			5	470	940	1400				5	590	1180	1770	2360		
6	55	620	1230	1600				6	670	1350	2020	2690			6	510	1010	1400				6	650	1290	1940	2500		
7	60	670	1330	1600				7	730	1470	2200	2930			7	540	1090	1400				7	700	1400	2110	2500		
8	66	730	1450	1600				8	800	1610	2410	2960			8	580	1170	1400				8	770	1530	2300	2500		
9	70	760	1530	1600				9	850	1700	2550	2960			9	610	1220	1400				9	810	1620	2430	2500		
10	80	850	1600					10	960	1920	2880	2960			10	670	1340	1400				10	910	1820	2500			
11	90	940	1600					11	1070	2130	2960				11	720	1400					11	1010	2010	2500			
12	100	1020	1600					12	1170	2340	2960				12	770	1400					12	1100	2190	2500			
13	110	1100	1600					13	1270	2540	2960				13	820	1400					13	1180	2370	2500			
14	120	1170	1600					14	1360	2730	2960				14	860	1400					14	1270	2500				
15	150	1360	1600					15	1630	2960					15	960	1400					15	1490	2500				
16	200	1600						16	2010	2960					16	1090	1400					16	1810	2500				
17	250	1600						17	2330	2960					17	1190	1400					17	2080	2500				
18	280	1600						18	2500	2960					18	1230	1400					18	2210	2500				

35 CLASS DIESEL LOCOMOTIVES

AIRBRAKE TRAINS >12.5 to 17 t/a								AIRBRAKE TRAINS >17 to 19 t/a							AIRBRAKE TRAINS >19 to 20 t/a							MOTOR CAR TRAINS, 8 to 12 t/a #						
CLASS : 35D								CLASS : 35D							CLASS : 35D							CLASS : 35D						
		(Number)							(Number)							(Number)							(Number)					
GC	GRAD.	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6
1	30	360	720	1080	1430	-	-	1	360	730	1090	1460	-	-	1	370	730	1100	1460	-	-	1	260	520	790	1050	-	-
2	35	430	850	1280	1710			2	430	870	1300	1740			2	440	870	1310	1750			2	310	630	940	1260		
3	40	490	990	1480	1980			3	500	1010	1510	2010			3	510	1010	1520	2030			3	370	730	1100	1460		
4	45	560	1120	1680	2240			4	570	1140	1710	2290			4	580	1150	1730	2300			4	410	830	1240	1660		
5	50	620	1250	1870	2500			5	640	1280	1920	2550			5	640	1290	1930	2570			5	460	930	1390	1850		
6	55	690	1380	2060	2750			6	700	1410	2110	2820			6	710	1420	2130	2840			6	510	1020	1530	2040		
7	60	750	1500	2250	3000			7	770	1540	2310	3080			7	780	1550	2330	3100			7	560	1110	1670	2220		
8	66	820	1650	2470	3290			8	850	1690	2540	3390			8	850	1710	2560	3420			8	610	1220	1830	2400		
9	70	870	1740	2610	3400			9	900	1790	2690	3590			9	910	1810	2720	3620			9	640	1290	1930	2400		
10	80	990	1980	2960	3400			10	1020	2040	3060	3800			10	1030	2060	3090	4000			10	730	1460	2180	2400		
11	90	1100	2200	3300	3400			11	1140	2280	3420	3800			11	1150	2310	3460	4000			11	810	1620	2400			
12	100	1210	2420	3400				12	1260	2510	3770	3800			12	1270	2540	3820	4000			12	880	1770	2400			
13	110	1310	2630	3400				13	1370	2740	3800				13	1390	2780	4000				13	960	1910	2400			
14	120	1420	2830	3400				14	1480	2960	3800				14	1500	3000	4000				14	1030	2050	2400			
15	150	1700	3400					15	1790	3590	3800				15	1820	3650	4000				15	1220	2400				
16	200	2120	3400					16	2260	3800					16	2310	4000					16	1490	2400				
17	250	2480	3400					17	2680	3800					17	2740	4000					17	1720	2400				
18	280	2680	3400					18	2900	3800					18	2980	4000					18	1840	2400				

RELEVANT TO VACUUM BRAKED TRAINS THE FOLLOWING RESTRICTIONS MAY NOT BE EXCEEDED:

a. Maximum train speed of 60 km/h.
b. Maximum average axle load per train of 18.5 ton per axle.
c. Maximum axle load per truck per axle of 20 ton.
d. Maximum train length of 40 trucks (160 axles).
e. Maximum train length on specific line as restricted in WTB.
f. Maximum axle load restriction on specific line.
g. Maximum total load of 2560 ton on category 1 lines where SASKOP-couplers are used.
h. Maximum number of locomotives in a consist is four (4).

Notes :

a. 'GC' Means Gradient Category.
b. Refer to point 5.9 below for Load Tables relevant to Passenger Trains.

This document is classified as 'Confidential' and as such is authorized and approved for general distribution within Transnet Freight Rail. (The document and/or any information contained herein may not be disclosed outside of TFR without prior permission first being obtained from the Principal Engineer (Train Design Technologies))

5.12 GENERAL LOAD TABLES FOR 35-800D LOCOMOTIVES

LOAD TABLES FOR 35-800 CLASS DIESEL LOCOMOTIVES (AURIZON)
AIRBRAKE - 50 WAGON TRAIN LENGTH MAXIMUM
VACUUM BRAKE - 40 WAGON TRAIN LENGTH MAXIMUM

TONNAGE RATINGS BY LINE PER LOCOMOTIVE

35-800 CLASS DIESEL LOCOMOTIVES (AURIZON)

VACUUM BRAKED TRAINS ≤ 10 TONS PER AXLE								VACUUM BRAKED TRAINS > 10 TONS PER AXLE						
CLASS : 35-800D								CLASS : 35-800D						
		(NUMBER)							(NUMBER)					
GC	GRAD.	1	2	3	4	5	6	GC	1	2	3	4	5	6
1	30	-	-	-	-	-	-	1	-	-	-	-	-	-
2	35							2						
3	40							3						
4	45							4						
5	50							5						
6	55							6						
7	60							7						
8	66	35-800 class locomotives are only airbrake equipped						8	35-800 class locomotives are only airbrake equipped					
9	70							9						
10	80							10						
11	90							11						
12	100							12						
13	110							13						
14	120							14						
15	150							15						
16	200							16						
17	250							17						
18	280							18						

AIRBRAKE TRAINS >2 to 7 t/a							AIRBRAKE TRAINS >7 to 12.5 t/a						
CLASS : 35-800D							CLASS : 35-800D						
	(NUMBER)							(NUMBER)					
GC	1	2	3	4	5	6	GC	1	2	3	4	5	6
1	420	850	1270	1400	-	-	1	500	990	1490	1990	-	-
2	490	980	1400				2	590	1170	1760	2340		
3	560	1110	1400				3	670	1340	2020	2500		
4	610	1230	1400				4	760	1510	2270	2500		
5	670	1340	1400				5	840	1680	2500			
6	720	1400					6	920	1840	2500			
7	770	1400					7	990	1990	2500			
8	830	1400					8	1080	2170	2500			
9	860	1400					9	1140	2290	2500			
10	950	1400					10	1280	2500				
11	1020	1400					11	1420	2500				
12	1090	1400					12	1550	2500				
13	1150	1400					13	1670	2500				
14	1210	1400					14	1780	2500				
15	1350	1400					15	2100	2500				
16	1400						16	2500					
17	1400						17	2500					
18	1400						18	2500					

35-800 CLASS DIESEL LOCOMOTIVES (AURIZON)

AIRBRAKE TRAINS >12.5 to 17 t/a								AIRBRAKE TRAINS >17 to 19 t/a						
CLASS : 35-800D								CLASS : 35-800D						
		(Number)							(Number)					
GC	GRAD.	1	2	3	4	5	6	GC	1	2	3	4	5	6
1	30	520	1030	1550	2070	-	-	1	520	1050	1570	2100	-	-
2	35	610	1220	1830	2450			2	620	1240	1860	2490		
3	40	700	1410	2110	2820			3	720	1440	2150	2870		
4	45	800	1590	2390	3190			4	810	1630	2440	3250		
5	50	890	1770	2660	3400			5	910	1810	2720	3620		
6	55	970	1950	2920	3400			6	1000	2000	2990	3800		
7	60	1060	2120	3180	3400			7	1090	2180	3270	3800		
8	66	1160	2330	3400				8	1200	2390	3590	3800		
9	70	1230	2460	3400				9	1270	2530	3800			
10	80	1390	2780	3400				10	1440	2880	3800			
11	90	1550	3100	3400				11	1600	3210	3800			
12	100	1700	3400	3400				12	1770	3530	3800			
13	110	1850	3400					13	1920	3800				
14	120	1990	3400					14	2080	3800				
15	150	2380	3400					15	2510	3800				
16	200	2970	3400					16	3170	3800				
17	250	3400						17	3750	3800				
18	280	3400						18	3800					

AIRBRAKE TRAINS >19 to 20 t/a							MOTOR CAR TRAINS, 8 to 12 t/a #						
CLASS : 35-800D							CLASS : 35-800D						
	(Number)							(Number)					
GC	1	2	3	4	5	6	GC	1	2	3	4	5	6
1	530	1050	1580	2100	-	-	1	380	760	1150	1530	-	-
2	620	1250	1870	2500			2	450	910	1360	1820		
3	720	1440	2170	2890			3	530	1050	1580	2100		
4	820	1640	2450	3270			4	590	1190	1780	2380		
5	910	1820	2740	3650			5	660	1320	1980	2400		
6	1010	2010	3020	4000			6	730	1450	2180	2400		
7	1100	2190	3290	4000			7	790	1580	2370	2400		
8	1210	2410	3620	4000			8	860	1730	2400			
9	1280	2550	3830	4000			9	910	1830	2400			
10	1450	2900	4000				10	1030	2060	2400			
11	1620	3250	4000				11	1140	2280	2400			
12	1790	3580	4000				12	1250	2400				
13	1950	3900	4000				13	1350	2400				
14	2110	4000					14	1450	2400				
15	2560	4000					15	1720	2400				
16	3230	4000					16	2100	2400				
17	3840	4000					17	2400					
18	4000						18	2400					

RELEVANT TO VACUUM BRAKED TRAINS THE FOLLOWING RESTRICTIONS MAY NOT BE EXCEEDED:

a. Maximum train speed of 60 km/h.
b. Maximum average axle load per train of 18.5 ton per axle.
c. Maximum axle load per truck per axle of 20 ton.
d. Maximum train length of 40 trucks (160 axles).
e. Maximum train length on specific line as restricted in WTB.
f. Maximum axle load restriction on specific line.
g. Maximum total load of 2560 ton on category 1 lines where SASKOP-couplers are used.
h. Maximum number of locomotives in a consist is four (4).

Notes :
a. 'GC' Means Gradient Category.
b. Refer to point 5.9 below for Load Tables relevant to Passenger Trains.

The class 35-800D locomotives are not authorized to MU with any other class TFR diesel locomotives as it is only possible to couple the brake pipe through. It must additionally be noted that the brake pipe needs to be coupled straight across and not under the coupler as is the current TFR regime, as this leads to the pipes uncoupling when negotiating a cross over or tight curve.

This document is classified as 'Confidential' and as such is authorized and approved for general distribution within Transnet Freight Rail. (The document and/or any information contained herein may not be disclosed outside of TFR without prior permission first being obtained from the Principal Engineer (Train Design Technologies))

5.13 GENERAL LOAD TABLES FOR 36D LOCOMOTIVES

LOAD TABLES FOR 36 CLASS DIESEL LOCOMOTIVES
AIRBRAKE - 50 WAGON TRAIN LENGTH MAXIMUM
VACUUM BRAKE - 40 WAGON TRAIN LENGTH MAXIMUM

TONNAGE RATINGS BY LINE PER LOCOMOTIVE

36 CLASS DIESEL LOCOMOTIVES

VACUUM BRAKED TRAINS ≤ 10 TONS PER AXLE								VACUUM BRAKED TRAINS > 10 TONS PER AXLE								AIRBRAKE TRAINS >2 to 7 t/a								AIRBRAKE TRAINS >7 to 12.5 t/a							
CLASS : 36D								CLASS : 36D								CLASS : 36D								CLASS : 36D							
		(NUMBER)							(NUMBER)								(NUMBER)								(NUMBER)						
GC	GRAD.	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6
1	30	270	550	820	1090	-	-	1	290	580	870	1160	-	-	1	240	480	720	960	-	-	1	280	560	850	1130	-	-	1	280	560	850	1130	-	-
2	35	320	650	970	1290			2	350	690	1040	1380			2	280	560	840	1120			2	340	670	1010	1340			2	340	670	1010	1340		
3	40	370	740	1120	1490			3	400	800	1200	1600			3	320	630	950	1270			3	390	770	1160	1550			3	390	770	1160	1550		
4	45	420	840	1260	1600			4	450	910	1360	1810			4	350	700	1060	1400			4	440	870	1310	1750			4	440	870	1310	1750		
5	50	460	930	1390	1600			5	500	1010	1510	2020			5	390	770	1160	1400			5	490	970	1460	1940			5	490	970	1460	1940		
6	55	510	1020	1520	1600			6	560	1110	1670	2220			6	420	830	1250	1400			6	530	1070	1600	2130			6	530	1070	1600	2130		
7	60	550	1100	1600				7	600	1210	1810	2420			7	450	890	1340	1400			7	580	1160	1740	2320			7	580	1160	1740	2320		
8	66	600	1200	1600				8	660	1330	1990	2650			8	480	960	1400				8	630	1270	1900	2500			8	630	1270	1900	2500		
9	70	630	1260	1600				9	700	1400	2100	2800			9	500	1000	1400				9	670	1330	2000	2500			9	670	1330	2000	2500		
10	80	710	1410	1600				10	790	1590	2380	2960			10	550	1100	1400				10	750	1500	2250	2500			10	750	1500	2250	2500		
11	90	780	1550	1600				11	880	1760	2650	2960			11	600	1190	1400				11	830	1660	2490	2500			11	830	1660	2490	2500		
12	100	840	1600					12	970	1930	2900	2960			12	640	1270	1400				12	910	1810	2500				12	910	1810	2500			
13	110	910	1600					13	1050	2100	2960				13	670	1350	1400				13	980	1960	2500				13	980	1960	2500			
14	120	960	1600					14	1130	2250	2960				14	710	1400					14	1050	2100	2500				14	1050	2100	2500			
15	150	1120	1600					15	1350	2690	2960				15	790	1400					15	1240	2470	2500				15	1240	2470	2500			
16	200	1340	1600					16	1660	2960					16	900	1400					16	1500	2500					16	1500	2500				
17	250	1520	1600					17	1930	2960					17	980	1400					17	1720	2500					17	1720	2500				
18	280	1600						18	2070	2960					18	1020	1400					18	1830	2500					18	1830	2500				

36 CLASS DIESEL LOCOMOTIVES

AIRBRAKE TRAINS >12.5 to 17 t/a								AIRBRAKE TRAINS >17 to 19 t/a								AIRBRAKE TRAINS >19 to 20 t/a								MOTOR CAR TRAINS, 8 to 12 t/a #							
CLASS : 36D								CLASS : 36D								CLASS : 36D								CLASS : 36D							
		(Number)							(Number)								(Number)								(Number)						
GC	GRAD.	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6
1	30	290	590	880	1170	-	-	1	300	600	890	1190	-	-	1	300	600	900	1200	-	-	1	210	430	640	860	-	-	1	210	430	640	860	-	-
2	35	350	700	1050	1400			2	360	710	1070	1430			2	360	720	1080	1430			2	260	510	770	1030			2	260	510	770	1030		
3	40	410	810	1220	1620			3	410	830	1240	1660			3	420	830	1250	1670			3	300	600	900	1200			3	300	600	900	1200		
4	45	460	920	1380	1840			4	470	940	1410	1880			4	470	950	1420	1890			4	340	680	1020	1360			4	340	680	1020	1360		
5	50	510	1030	1540	2060			5	530	1050	1580	2100			5	530	1060	1590	2120			5	380	760	1140	1520			5	380	760	1140	1520		
6	55	570	1130	1700	2270			6	580	1160	1740	2320			6	590	1170	1760	2340			6	420	840	1260	1680			6	420	840	1260	1680		
7	60	620	1240	1860	2470			7	630	1270	1900	2540			7	640	1280	1920	2560			7	460	910	1370	1830			7	460	910	1370	1830		
8	66	680	1360	2040	2720			8	700	1400	2100	2790			8	700	1410	2110	2820			8	500	1000	1500	2010			8	500	1000	1500	2010		
9	70	720	1440	2160	2880			9	740	1480	2220	2960			9	750	1490	2240	2990			9	530	1060	1590	2120			9	530	1060	1590	2120		
10	80	820	1630	2450	3260			10	840	1690	2530	3370			10	850	1700	2560	3410			10	600	1200	1800	2400			10	600	1200	1800	2400		
11	90	910	1820	2730	3400			11	940	1890	2830	3770			11	950	1910	2860	3810			11	670	1330	2000	2670			11	670	1330	2000	2670		
12	100	1000	2000	3000	3400			12	1040	2080	3120	3800			12	1050	2100	3160	4000			12	730	1460	2190	2920			12	730	1460	2190	2920		
13	110	1090	2170	3400				13	1130	2270	3400	3800			13	1150	2300	3450	4000			13	790	1580	2370	3160			13	790	1580	2370	3160		
14	120	1170	2340	3400				14	1220	2450	3670	3800			14	1240	2480	3730	4000			14	850	1700	2400				14	850	1700	2400			
15	150	1410	2820	3400				15	1480	2970	3800				15	1510	3020	4000				15	1010	2020	2400				15	1010	2020	2400			
16	200	1760	3400					16	1870	3750	3800				16	1910	3830	4000				16	1240	2400					16	1240	2400				
17	250	2060	3400					17	2220	3800					17	2270	4000					17	1430	2400					17	1430	2400				
18	280	2220	3400					18	2410	3800					18	2470	4000					18	1520	2400					18	1520	2400				

RELEVANT TO VACUUM BRAKED TRAINS THE FOLLOWING RESTRICTIONS MAY NOT BE EXCEEDED:

a. Maximum train speed of 60 km/h.
b. Maximum average axle load per train of 18.5 ton per axle.
c. Maximum axle load per truck per axle of 20 ton.
d. Maximum train length of 40 trucks (160 axles).
e. Maximum train length on specific line as restricted in WTB.
f. Maximum axle load restriction on specific line.
g. Maximum total load of 2560 ton on category 1 lines where SASKOP-couplers are used.
h. Maximum number of locomotives in a consist is four (4).

Notes :

a. 'GC' Means Gradient Category.
b. Refer to point 5.9 below for Load Tables relevant to Passenger Trains.

This document is classified as 'Confidential' and as such is authorized and approved for general distribution within Transnet Freight Rail. (The document and/or any information contained herein may not be disclosed outside of TFR without prior permission first being obtained from the Principal Engineer (Train Design Technologies))

5.14 GENERAL LOAD TABLES FOR 37D LOCOMOTIVES

LOAD TABLES FOR 37 CLASS AND NEXUS UPGRADED 34 CLASS DIESEL LOCOMOTIVES
AIRBRAKE - 50 WAGON TRAIN LENGTH MAXIMUM
VACUUM BRAKE - 40 WAGON TRAIN LENGTH MAXIMUM

TONNAGE RATINGS BY LINE PER LOCOMOTIVE

LOAD TABLES FOR 37 CLASS AND NEXUS UPGRADED 34 CLASS DIESEL LOCOMOTIVES

VACUUM BRAKED TRAINS ≤ 10 TONS PER AXLE								VACUUM BRAKED TRAINS > 10 TONS PER AXLE							AIRBRAKE TRAINS >2 to 7 t/a							AIRBRAKE TRAINS >7 to 12.5 t/a						
CLASS : 37D & NEXUS UPGRADED 34D								CLASS : 37D & NEXUS UPGRADED 34D							CLASS : 37D & NEXUS UPGRADED 34D							CLASS : 37D & NEXUS UPGRADED 34D						
		(NUMBER)							(NUMBER)							(NUMBER)							(NUMBER)					
GC	GRAD.	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6
1	30	510	1020	1530	1600	-	-	1	540	1080	1620	2150	-	-	1	450	890	1340	1400	-	-	1	520	1050	1570	2100	2500	-
2	35	600	1200	1600				2	640	1280	1920	2560			2	520	1040	1400				2	620	1240	1870	2490	2500	
3	40	690	1380	1600				3	740	1480	2220	2960			3	590	1180	1400				3	720	1430	2150	2500		
4	45	780	1550	1600				4	840	1670	2510	3350			4	650	1310	1400				4	810	1620	2430	2500		
5	50	860	1600					5	930	1870	2800	3730			5	710	1400					5	900	1800	2500			
6	55	940	1600					6	1030	2050	2960				6	770	1400					6	980	1970	2500			
7	60	1020	1600					7	1120	2230	2960				7	830	1400					7	1070	2140	2500			
8	66	1110	1600					8	1220	2450	2960				8	890	1400					8	1170	2340	2500			
9	70	1160	1600					9	1290	2590	2960				9	930	1400					9	1230	2460	2500			
10	80	1300	1600					10	1460	2930	2960				10	1020	1400					10	1390	2500				
11	90	1430	1600					11	1620	2960					11	1100	1400					11	1530	2500				
12	100	1550	1600					12	1780	2960					12	1170	1400					12	1670	2500				
13	110	1600						13	1930	2960					13	1240	1400					13	1800	2500				
14	120	1600						14	2080	2960					14	1300	1400					14	1930	2500				
15	150	1600						15	2480	2960					15	1400						15	2280	2500				
16	200	1600						16	2960						16	1400						16	2500					
17	250	1600						17	2960						17	1400						17	2500					
18	280	1600						18	2960						18	1400						18	2500					

37 CLASS DIESEL LOCOMOTIVES

AIRBRAKE TRAINS >12.5 to 17 t/a								AIRBRAKE TRAINS >17 to 19 t/a							AIRBRAKE TRAINS >19 to 20 t/a							MOTOR CAR TRAINS, 8 to 12 t/a #						
CLASS : 37D & NEXUS UPGRADED 34D								CLASS : 37D & NEXUS UPGRADED 34D							CLASS : 37D & NEXUS UPGRADED 34D							CLASS : 37D & NEXUS UPGRADED 34D						
		(Number)							(Number)							(Number)							(Number)					
GC	GRAD.	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6
1	30	550	1090	1640	2180	2730	3280	1	550	1110	1660	2220	2770	3320	1	560	1110	1670	2230	2780	3340	1	400	800	1200	1600	2000	2400
2	35	650	1300	1950	2600	3250	3400	2	660	1320	1980	2640	3310	3800	2	660	1330	1990	2660	3320	3990	2	480	960	1440	1910	2390	2400
3	40	750	1500	2260	3010	3400		3	770	1530	2300	3070	3800		3	770	1540	2310	3080	3850	4000	3	560	1110	1670	2220	2400	
4	45	850	1710	2560	3400			4	870	1740	2610	3480	3800		4	880	1750	2630	3500	4000		4	630	1260	1890	2400		
5	50	950	1900	2850	3400			5	970	1940	2920	3800			5	980	1960	2940	3920	4000		5	700	1410	2110	2400		
6	55	1050	2090	3140	3400			6	1070	2150	3220	3800			6	1080	2160	3240	4000			6	780	1550	2330	2400		
7	60	1140	2280	3400				7	1170	2340	3520	3800			7	1180	2360	3540	4000			7	850	1690	2400			
8	66	1250	2510	3400				8	1290	2580	3800				8	1300	2600	3900	4000			8	930	1850	2400			
9	70	1330	2650	3400				9	1370	2730	3800				9	1380	2760	4000				9	980	1960	2400			
10	80	1500	3010	3400				10	1550	3110	3800				10	1570	3140	4000				10	1110	2220	2400			
11	90	1680	3350	3400				11	1740	3470	3800				11	1760	3510	4000				11	1230	2400				
12	100	1840	3400					12	1910	3800					12	1940	3870	4000				12	1350	2400				
13	110	2000	3400					13	2090	3800					13	2110	4000					13	1460	2400				
14	120	2160	3400					14	2250	3800					14	2290	4000					14	1560	2400				
15	150	2590	3400					15	2730	3800					15	2780	4000					15	1860	2400				
16	200	3230	3400					16	3450	3800					16	3520	4000					16	2280	2400				
17	250	3400						17	3800						17	4000						17	2400					
18	280	3400						18	3800						18	4000						18	2400					

RELEVANT TO VACUUM BRAKED TRAINS THE FOLLOWING RESTRICTIONS MAY NOT BE EXCEEDED:

a. Maximum train speed of 60 km/h.
b. Maximum average axle load per train of 18.5 ton per axle.
c. Maximum axle load per truck per axle of 20 ton.
d. Maximum train length of 40 trucks (160 axles).
e. Maximum train length on specific line as restricted in WTB.
f. Maximum axle load restriction on specific line.
g. Maximum total load of 2560 ton on category 1 lines where SASKOP-couplers are used.
h. Maximum number of locomotives in a consist is four (4).

Notes :

a. 'GC' Means Gradient Category.
b. Refer to point 5.9 below for Load Tables relevant to Passenger Trains.

This document is classified as 'Confidential' and as such is authorized and approved for general distribution within Transnet Freight Rail. (The document and/or any information contained herein may not be disclosed outside of TFR without prior permission first being obtained from the Principal Engineer (Train Design Technologies))

5.15 GENERAL LOAD TABLES FOR 38 DIESEL / ELECTRIC LOCOMOTIVES

LOAD TABLES FOR 38 CLASS LOCOMOTIVES (DIESEL / ELECTRIC MODE)
AIRBRAKE - 50 WAGON TRAIN LENGTH MAXIMUM
VACUUM BRAKE - 40 WAGON TRAIN LENGTH MAXIMUM

TONNAGE RATINGS BY LINE PER LOCOMOTIVE

38 CLASS LOCOMOTIVES (DIESEL / ELECTRIC MODE)

VACUUM BRAKED TRAINS ≤ 10 TONS PER AXLE								VACUUM BRAKED TRAINS > 10 TONS PER AXLE								AIRBRAKE TRAINS >2 to 7 t/a								AIRBRAKE TRAINS >7 to 12.5 t/a							
CLASS : 38 (DIESEL / ELECTRIC MODE)								CLASS : 38 (DIESEL / ELECTRIC MODE)								CLASS : 38 (DIESEL / ELECTRIC MODE)								CLASS : 38 (DIESEL / ELECTRIC MODE)							
		(NUMBER)							(NUMBER)								(NUMBER)								(NUMBER)						
GC	GRAD.	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6
1	30	420	830	1250	1600	-	-	1	440	880	1320	1750	-	-	1	370	730	1100	1400	-	-	1	430	860	1280	1710	-	-	1	430	860	1280	1710	-	-
2	35	490	970	1460	1600			2	520	1030	1550	2070			2	420	850	1270	1400			2	500	1010	1510	2010			2	500	1010	1510	2010		
3	40	560	1110	1600				3	590	1190	1780	2380			3	480	950	1400				3	580	1150	1730	2310			3	580	1150	1730	2310		
4	45	620	1240	1600				4	670	1340	2010	2680			4	530	1050	1400				4	650	1290	1940	2500			4	650	1290	1940	2500		
5	50	690	1370	1600				5	740	1490	2230	2960			5	570	1150	1400				5	720	1430	2150	2500			5	720	1430	2150	2500		
6	55	750	1500	1600				6	810	1630	2440	2960			6	620	1240	1400				6	780	1570	2350	2500			6	780	1570	2350	2500		
7	60	810	1600					7	890	1770	2660	2960			7	660	1320	1400				7	850	1700	2500				7	850	1700	2500			
8	66	880	1600					8	970	1940	2900	2960			8	710	1400					8	920	1850	2500				8	920	1850	2500			
9	70	920	1600					9	1020	2040	2960				9	740	1400					9	970	1950	2500				9	970	1950	2500			
10	80	1030	1600					10	1150	2300	2960				10	810	1400					10	1090	2190	2500				10	1090	2190	2500			
11	90	1130	1600					11	1280	2560	2960				11	870	1400					11	1210	2410	2500				11	1210	2410	2500			
12	100	1220	1600					12	1400	2800	2960				12	930	1400					12	1310	2500					12	1310	2500				
13	110	1310	1600					13	1510	2960					13	980	1400					13	1420	2500					13	1420	2500				
14	120	1400	1600					14	1630	2960					14	1030	1400					14	1510	2500					14	1510	2500				
15	150	1600						15	1940	2960					15	1150	1400					15	1780	2500					15	1780	2500				
16	200	1600						16	2380	2960					16	1310	1400					16	2160	2500					16	2160	2500				
17	250	1600						17	2760	2960					17	1400						17	2470	2500					17	2470	2500				
18	280	1600						18	2960						18	1400						18	2500						18	2500					

38 CLASS LOCOMOTIVES (DIESEL / ELECTRIC MODE)

AIRBRAKE TRAINS >12.5 to 17 t/a								AIRBRAKE TRAINS >17 to 19 t/a								AIRBRAKE TRAINS >19 to 20 t/a								MOTOR CAR TRAINS, 8 to 12 t/a #							
CLASS : 38 (DIESEL / ELECTRIC MODE)								CLASS : 38 (DIESEL / ELECTRIC MODE)								CLASS : 38 (DIESEL / ELECTRIC MODE)								CLASS : 38 (DIESEL / ELECTRIC MODE)							
		(Number)							(Number)								(Number)								(Number)						
GC	GRAD.	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6
1	30	440	890	1330	1780	-	-	1	450	900	1350	1800	-	-	1	450	910	1360	1810	-	-	1	330	660	990	1320	-	-	1	330	660	990	1320	-	-
2	35	520	1050	1570	2100			2	530	1070	1600	2130			2	540	1070	1610	2140			2	390	780	1180	1570			2	390	780	1180	1570		
3	40	600	1210	1810	2410			3	610	1230	1840	2460			3	620	1240	1850	2470			3	450	900	1360	1810			3	450	900	1360	1810		
4	45	680	1360	2040	2720			4	690	1390	2080	2780			4	700	1400	2100	2800			4	510	1020	1530	2040			4	510	1020	1530	2040		
5	50	760	1510	2270	3030			5	770	1550	2320	3090			5	780	1560	2340	3120			5	570	1130	1700	2270			5	570	1130	1700	2270		
6	55	830	1660	2500	3330			6	850	1700	2550	3410			6	860	1720	2570	3430			6	620	1240	1870	2400			6	620	1240	1870	2400		
7	60	900	1810	2710	3400			7	930	1860	2780	3710			7	940	1870	2810	3740			7	680	1350	2030	2400			7	680	1350	2030	2400		
8	66	990	1980	2970	3400			8	1020	2040	3050	3800			8	1030	2050	3080	4000			8	740	1480	2220	2400			8	740	1480	2220	2400		
9	70	1050	2090	3140	3400			9	1080	2160	3230	3800			9	1090	2170	3260	4000			9	780	1560	2340	2400			9	780	1560	2340	2400		
10	80	1180	2370	3400				10	1220	2450	3670	3800			10	1240	2470	3710	4000			10	880	1760	2400				10	880	1760	2400			
11	90	1320	2630	3400				11	1360	2730	3800				11	1380	2760	4000				11	970	1950	2400				11	970	1950	2400			
12	100	1440	2890	3400				12	1500	3000	3800				12	1520	3040	4000				12	1060	2130	2400				12	1060	2130	2400			
13	110	1570	3140	3400				13	1630	3270	3800				13	1660	3310	4000				13	1150	2300	2400				13	1150	2300	2400			
14	120	1690	3400					14	1760	3530	3800				14	1790	3580	4000				14	1230	2400					14	1230	2400				
15	150	2020	3400					15	2130	3800					15	2170	4000					15	1460	2400					15	1460	2400				
16	200	2520	3400					16	2680	3800					16	2740	4000					16	1780	2400					16	1780	2400				
17	250	2940	3400					17	3170	3800					17	3250	4000					17	2050	2400					17	2050	2400				
18	280	3170	3400					18	3440	3800					18	3530	4000					18	2190	2400					18	2190	2400				

RELEVANT TO VACUUM BRAKED TRAINS THE FOLLOWING RESTRICTIONS MAY NOT BE EXCEEDED:

a. Maximum train speed of 60 km/h.
b. Maximum average axle load per train of 18.5 ton per axle.
c. Maximum axle load per truck per axle of 20 ton.
d. Maximum train length of 40 trucks (160 axles).
e. Maximum train length on specific line as restricted in WTB.
f. Maximum axle load restriction on specific line.
g. Maximum total load of 2560 ton on category 1 lines where SASKOP-couplers are used.
h. Maximum number of locomotives in a consist is four (4).

Notes :

a. 'GC' Means Gradient Category.
b. Refer to point 5.9 below for Load Tables relevant to Passenger Trains.
c. The tables for Class 38 locomotives apply for diesel as well as electric mode.

This document is classified as 'Confidential' and as such is authorized and approved for general distribution within Transnet Freight Rail. (The document and/or any information contained herein may not be disclosed outside of TFR without prior permission first being obtained from the Principal Engineer (Train Design Technologies))

5.16 GENERAL LOAD TABLES FOR 39-000 DIESEL LOCOMOTIVES

LOAD TABLES FOR 39-000 CLASS DIESEL LOCOMOTIVES
AIRBRAKE - 50 WAGON TRAIN LENGTH MAXIMUM
VACUUM BRAKE - 40 WAGON TRAIN LENGTH MAXIMUM

TONNAGE RATINGS BY LINE PER LOCOMOTIVE
39-000 CLASS DIESEL LOCOMOTIVES

VACUUM BRAKED TRAINS ≤ 10 TONS PER AXLE								VACUUM BRAKED TRAINS > 10 TONS PER AXLE							AIRBRAKE TRAINS >2 to 7 t/a							AIRBRAKE TRAINS >7 to 12.5 t/a						
CLASS : 39-000 DIESEL LOCOMOTIVES								CLASS : 39-000 DIESEL LOCOMOTIVES							CLASS : 39-000 DIESEL LOCOMOTIVES							CLASS : 39-000 DIESEL LOCOMOTIVES						
		(NUMBER)							(NUMBER)							(NUMBER)							(NUMBER)					
GC	GRAD.	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6
1	30	620	1250	1600	-	-	-	1	660	1320	1980	2630	-	-	1	550	1100	1400	-	-	-	1	640	1280	1930	2500	-	-
2	35	730	1460	1600				2	780	1560	2340	2960			2	640	1270	1400				2	760	1510	2270	2500		
3	40	840	1600					3	900	1790	2690	2960			3	720	1400					3	870	1740	2500			
4	45	940	1600					4	1010	2020	2960				4	790	1400					4	980	1950	2500			
5	50	1040	1600					5	1120	2250	2960				5	870	1400					5	1080	2160	2500			
6	55	1130	1600					6	1230	2470	2960				6	930	1400					6	1180	2370	2500			
7	60	1220	1600					7	1340	2680	2960				7	1000	1400					7	1280	2500				
8	66	1330	1600					8	1470	2930	2960				8	1070	1400					8	1400	2500				
9	70	1400	1600					9	1550	2960					9	1120	1400					9	1480	2500				
10	80	1560	1600					10	1750	2960					10	1220	1400					10	1660	2500				
11	90	1600						11	1940	2960					11	1320	1400					11	1830	2500				
12	100	1600						12	2120	2960					12	1400						12	2000	2500				
13	110	1600						13	2300	2960					13	1400						13	2150	2500				
14	120	1600						14	2470	2960					14	1400						14	2300	2500				
15	150	1600						15	2950	2960					15	1400						15	2500					
16	200	1600						16	2960						16	1400						16	2500					
17	250	1600						17	2960						17	1400						17	2500					
18	280	1600						18	2960						18	1400						18	2500					

39-000 CLASS DIESEL LOCOMOTIVES

AIRBRAKE TRAINS >12.5 to 17 t/a								AIRBRAKE TRAINS >17 to 19 t/a							AIRBRAKE TRAINS >19 to 20 t/a							MOTOR CAR TRAINS, 8 to 12 t/a #						
CLASS : 39-000 DIESEL LOCOMOTIVES								CLASS : 39-000 DIESEL LOCOMOTIVES							CLASS : 39-000 DIESEL LOCOMOTIVES							CLASS : 39-000 DIESEL LOCOMOTIVES						
		(Number)							(Number)							(Number)							(Number)					
GC	GRAD.	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6
1	30	670	1330	2000	2670	3330	3400	1	680	1350	2030	2710	3380	3800	1	680	1360	2040	2720	3400	4000	1	490	990	1480	1980	2400	-
2	35	790	1580	2370	3160	3400		2	800	1610	2410	3210	3800		2	810	1610	2420	3230	4000		2	590	1170	1760	2350	2400	
3	40	910	1820	2730	3400			3	930	1850	2780	3710	3800		3	930	1860	2800	3730	4000		3	680	1360	2040	2400		
4	45	1030	2060	3090	3400			4	1050	2100	3150	3800			4	1060	2110	3170	4000			4	770	1530	2300	2400		
5	50	1140	2290	3400				5	1170	2340	3510	3800			5	1180	2360	3530	4000			5	850	1710	2400			
6	55	1260	2520	3400				6	1290	2580	3800				6	1300	2600	3900	4000			6	940	1880	2400			
7	60	1370	2740	3400				7	1410	2810	3800				7	1420	2830	4000				7	1020	2040	2400			
8	66	1500	3000	3400				8	1540	3090	3800				8	1560	3110	4000				8	1120	2230	2400			
9	70	1590	3180	3400				9	1630	3270	3800				9	1650	3300	4000				9	1180	2360	2400			
10	80	1800	3400					10	1860	3710	3800				10	1880	3750	4000				10	1330	2400				
11	90	2000	3400					11	2070	3800					11	2100	4000					11	1470	2400				
12	100	2200	3400					12	2280	3800					12	2310	4000					12	1610	2400				
13	110	2380	3400					13	2490	3800					13	2520	4000					13	1740	2400				
14	120	2570	3400					14	2680	3800					14	2720	4000					14	1870	2400				
15	150	3080	3400					15	3250	3800					15	3300	4000					15	2220	2400				
16	200	3400						16	3800						16	4000						16	2400					
17	250	3400						17	3800						17	4000						17	2400					
18	280	3400						18	3800						18	4000						18	2400					

RELEVANT TO VACUUM BRAKED TRAINS THE FOLLOWING RESTRICTIONS MAY NOT BE EXCEEDED:

a. Maximum train speed of 60 km/h.
b. Maximum average axle load per train of 18.5 ton per axle.
c. Maximum axle load per truck per axle of 20 ton.
d. Maximum train length of 40 trucks (160 axles).
e. Maximum train length on specific line as restricted in WTB.
f. Maximum axle load restriction on specific line.
g. Maximum total load of 2560 ton on category 1 lines where SASKOP-couplers are used.
h. Maximum number of locomotives in a consist is four (4).

Notes :

a. 'GC' Means Gradient Category.
b. Refer to point 5.9 below for Load Tables relevant to Passenger Trains.

This document is classified as 'Confidential' and as such is authorized and approved for general distribution within Transnet Freight Rail. (The document and/or any information contained herein may not be disclosed outside of TFR without prior permission first being obtained from the Principal Engineer (Train Design Technologies))

5.17 GENERAL LOAD TABLES FOR 39-200 DIESEL LOCOMOTIVES

LOAD TABLES FOR 39-200 CLASS DIESEL LOCOMOTIVES
AIRBRAKE - 50 WAGON TRAIN LENGTH MAXIMUM
VACUUM BRAKE - 40 WAGON TRAIN LENGTH MAXIMUM

TONNAGE RATINGS BY LINE PER LOCOMOTIVE

39-200 CLASS DIESEL LOCOMOTIVES

VACUUM BRAKED TRAINS ≤ 10 TONS PER AXLE								VACUUM BRAKED TRAINS > 10 TONS PER AXLE							AIRBRAKE TRAINS >2 to 7 t/a							AIRBRAKE TRAINS >7 to 12.5 t/a						
CLASS : 39-200-200 DIESEL LOCOMOTIVES								CLASS : 39-200-200 DIESEL LOCOMOTIVES							CLASS : 39-200-200 DIESEL LOCOMOTIVES							CLASS : 39-200-200 DIESEL LOCOMOTIVES						
		(NUMBER)							(NUMBER)							(NUMBER)							(NUMBER)					
GC	GRAD.	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6
1	30	620	1240	1600	-	-	-	1	660	1310	1970	2630	-	-	1	550	1100	1400	-	-	-	1	640	1280	1920	2500	-	-
2	35	730	1460	1600				2	780	1560	2330	2960			2	630	1270					2	760	1510	2270	2500		
3	40	840	1600					3	900	1790	2690				3	720	1400					3	870	1730	2500			
4	45	940	1600					4	1010	2020	2960				4	790	1400					4	980	1950	2500			
5	50	1030	1600					5	1120	2240	2960				5	860	1400					5	1080	2160	2500			
6	55	1130	1600					6	1230	2460	2960				6	930	1400					6	1180	2370	2500			
7	60	1220	1600					7	1340	2680	2960				7	1000	1400					7	1280	2500				
8	66	1330	1600					8	1470	2930	2960				8	1070	1400					8	1400	2500				
9	70	1390	1600					9	1550	2960					9	1120	1400					9	1480	2500				
10	80	1560	1600					10	1750	2960					10	1220	1400					10	1660	2500				
11	90	1600						11	1940	2960					11	1320	1400					11	1830	2500				
12	100	1600						12	2120	2960					12	1400						12	1990	2500				
13	110	1600						13	2300	2960					13	1400						13	2150	2500				
14	120	1600						14	2470	2960					14	1400						14	2300	2500				
15	150	1600						15	2940	2960					15	1400						15	2500					
16	200	1600						16	2960						16	1400						16	2500					
17	250	1600						17	2960						17	1400						17	2500					
18	280	1600						18	2960						18	1400						18	2500					

39-200 CLASS DIESEL LOCOMOTIVES

AIRBRAKE TRAINS >12.5 to 17 t/a								AIRBRAKE TRAINS >17 to 19 t/a							AIRBRAKE TRAINS >19 to 20 t/a							MOTOR CAR TRAINS, 8 to 12 t/a #						
CLASS : 39-200-200 DIESEL LOCOMOTIVES								CLASS : 39-200-200 DIESEL LOCOMOTIVES							CLASS : 39-200-200 DIESEL LOCOMOTIVES							CLASS : 39-200-200 DIESEL LOCOMOTIVES						
		(Number)							(Number)							(Number)							(Number)					
GC	GRAD.	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6
1	30	670	1330	2000	2660	3330	-	1	680	1350	2030	2700	3380	-	1	680	1360	2040	2710	3390	4000	1	490	990	1480	1970	2400	-
2	35	790	1580	2370	3160	3400		2	800	1600	2410	3210	3800		2	810	1610	2420	3220	4000		2	590	1170	1760	2350	2400	
3	40	910	1820	2730	3400			3	930	1850	2780	3700	3800		3	930	1860	2790	3730	4000		3	680	1360	2030	2400		
4	45	1030	2060	3080	3400			4	1050	2100	3150	3800			4	1060	2110	3170	4000			4	770	1530	2300	2400		
5	50	1140	2290	3400				5	1170	2340	3510	3800			5	1180	2350	3530	4000			5	850	1710	2400			
6	55	1260	2520	3400				6	1290	2580	3800				6	1300	2590	3890	4000			6	940	1870	2400			
7	60	1370	2740	3400				7	1400	2810	3800				7	1420	2830	4000				7	1020	2040	2400			
8	66	1500	3000	3400				8	1540	3090	3800				8	1560	3110	4000				8	1120	2230	2400			
9	70	1590	3170	3400				9	1630	3270	3800				9	1650	3300	4000				9	1180	2360	2400			
10	80	1800	3400					10	1860	3710	3800				10	1870	3750	4000				10	1330	2400				
11	90	2000	3400					11	2070	3800					11	2090	4000					11	1470	2400				
12	100	2190	3400					12	2280	3800					12	2310	4000					12	1610	2400				
13	110	2380	3400					13	2480	3800					13	2520	4000					13	1740	2400				
14	120	2570	3400					14	2680	3800					14	2720	4000					14	1870	2400				
15	150	3080	3400					15	3240	3800					15	3300	4000					15	2210	2400				
16	200	3400						16	3800						16	4000						16	2400					
17	250	3400						17	3800						17	4000						17	2400					
18	280	3400						18	3800						18	4000						18	2400					

RELEVANT TO VACUUM BRAKED TRAINS THE FOLLOWING RESTRICTIONS MAY NOT BE EXCEEDED:

a. Maximum train speed of 60 km/h.
b. Maximum average axle load per train of 18.5 ton per axle.
c. Maximum axle load per truck per axle of 20 ton.
d. Maximum train length of 40 trucks (160 axles).
e. Maximum train length on specific line as restricted in WTB.
f. Maximum axle load restriction on specific line.
g. Maximum total load of 2560 ton on category 1 lines where SASKOP-couplers are used.
h. Maximum number of locomotives in a consist is four (4).

Notes :

a. 'GC' Means Gradient Category.
b. Refer to point 5.9 below for Load Tables relevant to Passenger Trains.

This document is classified as 'Confidential' and as such is authorized and approved for general distribution within Transnet Freight Rail. (The document and/or any information contained herein may not be disclosed outside of TFR without prior permission first being obtained from the Principal Engineer (Train Design Technologies))

5.18 GENERAL LOAD TABLES FOR CLASS 43 DIESEL LOCOMOTIVES

LOAD TABLES FOR 43 CLASS DIESEL LOCOMOTIVES
AIRBRAKE - 50 WAGON TRAIN LENGTH MAXIMUM
VACUUM BRAKE - 40 WAGON TRAIN LENGTH MAXIMUM

TONNAGE RATINGS BY LINE PER LOCOMOTIVE

43 CLASS DIESEL LOCOMOTIVES

VACUUM BRAKED TRAINS ≤ 10 TONS PER AXLE								VACUUM BRAKED TRAINS > 10 TONS PER AXLE							AIRBRAKE TRAINS >2 to 7 t/a							AIRBRAKE TRAINS >7 to 12.5 t/a						
CLASS : 43D								CLASS : 43D							CLASS : 43D							CLASS : 43D						
		(NUMBER)							(NUMBER)							(NUMBER)							(NUMBER)					
GC	GRAD.	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6
1	30	820	1600	-	-	-	-	1	870	1740	2960	-	-	-	1	730	1400	-	-	-	-	1	850	1690	2500	-	-	-
2	35	960	1600					2	1020	2040	2960				2	840	1400					2	990	1990	2500			
3	40	1090	1600					3	1170	2340	2960				3	940	1400					3	1130	2270	2500			
4	45	1220	1600					4	1320	2630	2960				4	1040	1400					4	1270	2500				
5	50	1350	1600					5	1460	2920	2960				5	1130	1400					5	1410	2500				
6	55	1470	1600					6	1600	2960					6	1220	1400					6	1540	2500				
7	60	1580	1600					7	1730	2960					7	1300	1400					7	1660	2500				
8	66	1600						8	1890	2960					8	1390	1400					8	1810	2500				
9	70	1600						9	2000	2960					9	1400						9	1910	2500				
10	80	1600						10	2250	2960					10	1400						10	2140	2500				
11	90	1600						11	2500	2960					11	1400						11	2360	2500				
12	100	1600						12	2730	2960					12	1400						12	2500					
13	110	1600						13	2950	2960					13	1400						13	2500					
14	120	1600						14	2960						14	1400						14	2500					
15	150	1600						15	2960						15	1400						15	2500					
16	200	1600						16	2960						16	1400						16	2500					
17	250	1600						17	2960						17	1400						17	2500					
18	280	1600						18	2960						18	1400						18	2500					

43 CLASS DIESEL LOCOMOTIVES

AIRBRAKE TRAINS >12.5 to 17 t/a								AIRBRAKE TRAINS >17 to 19 t/a							AIRBRAKE TRAINS >19 to 20 t/a							MOTOR CAR TRAINS, 8 to 12 t/a #						
CLASS : 43D								CLASS : 43D							CLASS : 43D							CLASS : 43D						
		(Number)							(Number)							(Number)							(Number)					
GC	GRAD.	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6
1	30	880	1760	2640	3400	-	-	1	890	1780	2670	3570	3800	-	1	900	1790	2690	3580	4000	-	1	660	1320	1980	2400	-	-
2	35	1030	2070	3100	3400			2	1050	2100	3150	3800			2	1060	2110	3170	4000			2	780	1560	2330	2400		
3	40	1190	2380	3400				3	1210	2420	3630	3800			3	1220	2430	3650	4000			3	890	1790	2400			
4	45	1340	2680	3400				4	1360	2730	3800				4	1370	2750	4000				4	1010	2010	2400			
5	50	1490	2970	3400				5	1520	3040	3800				5	1530	3060	4000				5	1120	2230	2400			
6	55	1630	3260	3400				6	1670	3340	3800				6	1680	3360	4000				6	1220	2400				
7	60	1770	3400					7	1820	3630	3800				7	1830	3660	4000				7	1330	2400				
8	66	1940	3400					8	1990	3800					8	2010	4000					8	1450	2400				
9	70	2050	3400					9	2110	3800					9	2130	4000					9	1530	2400				
10	80	2310	3400					10	2390	3800					10	2410	4000					10	1720	2400				
11	90	2570	3400					11	2660	3800					11	2690	4000					11	1900	2400				
12	100	2820	3400					12	2930	3800					12	2960	4000					12	2080	2400				
13	110	3060	3400					13	3190	3800					13	3230	4000					13	2240	2400				
14	120	3290	3400					14	3440	3800					14	3490	4000					14	2400					
15	150	3400						15	3800						15	4000						15	2400					
16	200	3400						16	3800						16	4000						16	2400					
17	250	3400						17	3800						17	4000						17	2400					
18	280	3400						18	3800						18	4000						18	2400					

RELEVANT TO VACUUM BRAKED TRAINS THE FOLLOWING RESTRICTIONS MAY NOT BE EXCEEDED:

a. Maximum train speed of 60 km/h.
b. Maximum average axle load per train of 18.5 ton per axle.
c. Maximum axle load per truck per axle of 20 ton.
d. Maximum train length of 40 trucks (160 axles).
e. Maximum train length on specific line as restricted in WTB.
f. Maximum axle load restriction on specific line.
g. Maximum total load of 2560 ton on category 1 lines where SASKOP-couplers are used.
h. Maximum number of locomotives in a consist is four (4).

Notes : a. 'GC' Means Gradient Category.

This document is classified as 'Confidential' and as such is authorized and approved for general distribution within Transnet Freight Rail. (The document and/or any information contained herein may not be disclosed outside of TFR without prior permission first being obtained from the Principal Engineer (Train Design Technologies))

5.19 GENERAL LOAD TABLES FOR CLASS 44 DIESEL LOCOMOTIVES

LOAD TABLES FOR 44 CLASS DIESEL LOCOMOTIVES
AIRBRAKE - 50 WAGON TRAIN LENGTH MAXIMUM
VACUUM BRAKE - 40 WAGON TRAIN LENGTH MAXIMUM

TONNAGE RATINGS BY LINE PER LOCOMOTIVE

44 CLASS DIESEL LOCOMOTIVES

VACUUM BRAKED TRAINS ≤ 10 TONS PER AXLE								VACUUM BRAKED TRAINS > 10 TONS PER AXLE							AIRBRAKE TRAINS >2 to 7 t/a							AIRBRAKE TRAINS >7 to 12.5 t/a						
CLASS : 44D								CLASS : 44D							CLASS : 44D							CLASS : 44D						
		(NUMBER)							(NUMBER)							(NUMBER)							(NUMBER)					
GC	GRAD.	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6
1	30	910	1600	-	-	-	-	1	960	1910	2960	-	-	-	1	800	1400	-	-	-	-	1	930	1870	2500	-	-	-
2	35	1060	1600					2	1120	2250	2960				2	920	1400					2	1090	2190	2500			
3	40	1200	1600					3	1280	2570	2960				3	1040	1400					3	1250	2490	2500			
4	45	1340	1600					4	1440	2890	2960				4	1140	1400					4	1400	2500				
5	50	1480	1600					5	1600	2960					5	1240	1400					5	1540	2500				
6	55	1600						6	1750	2960					6	1340	1400					6	1680	2500				
7	60	1600						7	1900	2960					7	1400						7	1820	2500				
8	66	1600						8	2070	2960					8	1400						8	1980	2500				
9	70	1600						9	2190	2960					9	1400						9	2090	2500				
10	80	1600						10	2460	2960					10	1400						10	2340	2500				
11	90	1600						11	2730	2960					11	1400						11	2500					
12	100	1600						12	2960						12	1400						12	2500					
13	110	1600						13	2960						13	1400						13	2500					
14	120	1600						14	2960						14	1400						14	2500					
15	150	1600						15	2960						15	1400						15	2500					
16	200	1600						16	2960						16	1400						16	2500					
17	250	1600						17	2960						17	1400						17	2500					
18	280	1600						18	2960						18	1400						18	2500					

44 CLASS DIESEL LOCOMOTIVES

AIRBRAKE TRAINS >12.5 to 17 t/a								AIRBRAKE TRAINS >17 to 19 t/a							AIRBRAKE TRAINS >19 to 20 t/a							MOTOR CAR TRAINS, 8 to 12 t/a #						
CLASS : 44D								CLASS : 44D							CLASS : 44D							CLASS : 44D						
		(Number)							(Number)							(Number)							(Number)					
GC	GRAD.	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6
1	30	970	1940	2900	3400	-	-	1	980	1960	2940	3800	-	-	1	990	1970	2960	3940	4000	-	1	730	1460	2190	2400	-	-
2	35	1140	2280	3400				2	1160	2310	3470	3800			2	1160	2320	3490	4000			2	860	1720	2400			
3	40	1300	2610	3400				3	1330	2660	3800				3	1340	2670	4000				3	980	1970	2400			
4	45	1470	2940	3400				4	1500	2990	3800				4	1510	3010	4000				4	1110	2210	2400			
5	50	1630	3260	3400				5	1660	3330	3800				5	1670	3350	4000				5	1230	2400				
6	55	1790	3400					6	1830	3660	3800				6	1840	3680	4000				6	1340	2400				
7	60	1940	3400					7	1990	3800					7	2000	4000					7	1460	2400				
8	66	2120	3400					8	2180	3800					8	2200	4000					8	1590	2400				
9	70	2240	3400					9	2310	3800					9	2330	4000					9	1680	2400				
10	80	2530	3400					10	2610	3800					10	2640	4000					10	1880	2400				
11	90	2810	3400					11	2910	3800					11	2940	4000					11	2080	2400				
12	100	3080	3400					12	3200	3800					12	3240	4000					12	2270	2400				
13	110	3340	3400					13	3480	3800					13	3530	4000					13	2400					
14	120	3400						14	3750	3800					14	3810	4000					14	2400					
15	150	3400						15	3800						15	4000						15	2400					
16	200	3400						16	3800						16	4000						16	2400					
17	250	3400						17	3800						17	4000						17	2400					
18	280	3400						18	3800						18	4000						18	2400					

RELEVANT TO VACUUM BRAKED TRAINS THE FOLLOWING RESTRICTIONS MAY NOT BE EXCEEDED:

a. Maximum train speed of 60 km/h.
b. Maximum average axle load per train of 18.5 ton per axle.
c. Maximum axle load per truck per axle of 20 ton.
d. Maximum train length of 40 trucks (160 axles).
e. Maximum train length on specific line as restricted in WTB.
f. Maximum axle load restriction on specific line.
g. Maximum total load of 2560 ton on category 1 lines where SASKOP-couplers are used.
h. Maximum number of locomotives in a consist is four (4).

Notes : a. 'GC' Means Gradient Category.

This document is classified as 'Confidential' and as such is authorized and approved for general distribution within Transnet Freight Rail. (The document and/or any information contained herein may not be disclosed outside of TFR without prior permission first being obtained from the Principal Engineer (Train Design Technologies))

5.20 GENERAL LOAD TABLES FOR CLASS 45 DIESEL LOCOMOTIVES

LOAD TABLES FOR 45 CLASS DIESEL LOCOMOTIVES
AIRBRAKE - 50 WAGON TRAIN LENGTH MAXIMUM
VACUUM BRAKE - 40 WAGON TRAIN LENGTH MAXIMUM

TONNAGE RATINGS BY LINE PER LOCOMOTIVE

45 CLASS DIESEL LOCOMOTIVES

VACUUM BRAKED TRAINS ≤ 10 TONS PER AXLE								VACUUM BRAKED TRAINS > 10 TONS PER AXLE							AIRBRAKE TRAINS >2 to 7 t/a							AIRBRAKE TRAINS >7 to 12.5 t/a						
CLASS : 45D								CLASS : 45D							CLASS : 45D							CLASS : 45D						
		(NUMBER)							(NUMBER)							(NUMBER)							(NUMBER)					
GC	GRAD.	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6
1	30	910	1600	-	-	-	-	1	960	1910	2960	-	-	-	1	800	1400	-	-	-	-	1	930	1870	2500	-	-	-
2	35	1060	1600					2	1120	2250	2960				2	920	1400					2	1090	2190	2500			
3	40	1200	1600					3	1280	2570	2960				3	1040	1400					3	1250	2490	2500			
4	45	1340	1600					4	1440	2890	2960				4	1140	1400					4	1400	2500				
5	50	1480	1600					5	1600	2960					5	1240	1400					5	1540	2500				
6	55	1600						6	1750	2960					6	1340	1400					6	1680	2500				
7	60	1600						7	1900	2960					7	1400						7	1820	2500				
8	66	1600						8	2070	2960					8	1400						8	1980	2500				
9	70	1600						9	2190	2960					9	1400						9	2090	2500				
10	80	1600						10	2460	2960					10	1400						10	2340	2500				
11	90	1600						11	2730	2960					11	1400						11	2500					
12	100	1600						12	2960						12	1400						12	2500					
13	110	1600						13	2960						13	1400						13	2500					
14	120	1600						14	2960						14	1400						14	2500					
15	150	1600						15	2960						15	1400						15	2500					
16	200	1600						16	2960						16	1400						16	2500					
17	250	1600						17	2960						17	1400						17	2500					
18	280	1600						18	2960						18	1400						18	2500					

45 CLASS DIESEL LOCOMOTIVES

AIRBRAKE TRAINS >12.5 to 17 t/a								AIRBRAKE TRAINS >17 to 19 t/a							AIRBRAKE TRAINS >19 to 20 t/a							MOTOR CAR TRAINS, 8 to 12 t/a #						
CLASS : 45D								CLASS : 45D							CLASS : 45D							CLASS : 45D						
		(Number)							(Number)							(Number)							(Number)					
GC	GRAD.	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6
1	30	970	1940	2900	3400	-	-	1	980	1960	2940	3800	-	-	1	990	1970	2960	3940	4000	-	1	730	1460	2190	2400	-	-
2	35	1140	2280	3400				2	1160	2310	3470	3800			2	1160	2320	3490	4000			2	860	1720	2400			
3	40	1300	2610	3400				3	1330	2660	3800				3	1340	2670	4000				3	980	1970	2400			
4	45	1470	2940	3400				4	1500	2990	3800				4	1510	3010	4000				4	1110	2210	2400			
5	50	1630	3260	3400				5	1660	3330	3800				5	1670	3350	4000				5	1230	2400				
6	55	1790	3400					6	1830	3660	3800				6	1840	3680	4000				6	1340	2400				
7	60	1940	3400					7	1990	3800					7	2000	4000					7	1460	2400				
8	66	2120	3400					8	2180	3800					8	2200	4000					8	1590	2400				
9	70	2240	3400					9	2310	3800					9	2330	4000					9	1680	2400				
10	80	2530	3400					10	2610	3800					10	2640	4000					10	1880	2400				
11	90	2810	3400					11	2910	3800					11	2940	4000					11	2080	2400				
12	100	3080	3400					12	3200	3800					12	3240	4000					12	2270	2400				
13	110	3340	3400					13	3480	3800					13	3530	4000					13	2400					
14	120	3400						14	3750	3800					14	3810	4000					14	2400					
15	150	3400						15	3800						15	4000						15	2400					
16	200	3400						16	3800						16	4000						16	2400					
17	250	3400						17	3800						17	4000						17	2400					
18	280	3400						18	3800						18	4000						18	2400					

RELEVANT TO VACUUM BRAKED TRAINS THE FOLLOWING RESTRICTIONS MAY NOT BE EXCEEDED:

a. Maximum train speed of 60 km/h.
b. Maximum average axle load per train of 18.5 ton per axle.
c. Maximum axle load per truck per axle of 20 ton.
d. Maximum train length of 40 trucks (160 axles).
e. Maximum train length on specific line as restricted in WTB.
f. Maximum axle load restriction on specific line.
g. Maximum total load of 2560 ton on category 1 lines where SASKOP-couplers are used.

Notes : a. 'GC' Means Gradient Category.

This document is classified as 'Confidential' and as such is authorized and approved for general distribution within Transnet Freight Rail. (The document and/or any information contained herein may not be disclosed outside of TFR without prior permission first being obtained from the Principal Engineer (Train Design Technologies))

5.21 GENERAL LOAD TABLES FOR CLASS TAL 2000 DIESEL LOCOMOTIVES

LOAD TABLES FOR TAL 2000 DIESEL LOCOMOTIVES
TRANS AFRICA LOCOMOTIVES
AIRBRAKE - 50 WAGON TRAIN LENGTH MAXIMUM - VACUUM BRAKE - 40 WAGON TRAIN LENGTH MAXIMUM

TONNAGE RATINGS BY LINE PER LOCOMOTIVE

TAL 2000 CLASS DIESEL LOCOMOTIVES

VACUUM BRAKED TRAINS ≤ 10 TONS PER AXLE								VACUUM BRAKED TRAINS > 10 TONS PER AXLE							AIRBRAKE TRAINS >2 to 7 t/a							AIRBRAKE TRAINS >7 to 12.5 t/a						
CLASS : TAL 2000								CLASS : TAL 2000							CLASS : TAL 2000							CLASS : TAL 2000						
		(NUMBER)							(NUMBER)							(NUMBER)							(NUMBER)					
GC	GRAD.	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6
1	30	520	1040	1560	1600	-	-	1	550	1100	1650	2200	-	-	1	460	920	1380	1400	-	-	1	540	1070	1610	2150	-	-
2	35	610	1220	1600				2	650	1300	1950	2600			2	530	1060	1400				2	630	1260	1900	2500		
3	40	700	1400	1600				3	750	1490	2240	2960			3	600	1200	1400				3	720	1450	2170	2500		
4	45	780	1560	1600				4	840	1680	2520	2960			4	660	1320	1400				4	810	1630	2440	2500		
5	50	860	1600					5	930	1870	2800	2960			5	720	1400					5	900	1800	2500			
6	55	940	1600					6	1030	2050	2960				6	780	1400					6	990	1970	2500			
7	60	1020	1600					7	1110	2230	2960				7	830	1400					7	1070	2140	2500			
8	66	1100	1600					8	1220	2440	2960				8	890	1400					8	1160	2330	2500			
9	70	1160	1600					9	1290	2570	2960				9	930	1400					9	1230	2450	2500			
10	80	1290	1600					10	1450	2900	2960				10	1020	1400					10	1380	2500				
11	90	1420	1600					11	1610	2960					11	1100	1400					11	1520	2500				
12	100	1540	1600					12	1760	2960					12	1170						12	1650	2500				
13	110	1600						13	1910	2960					13	1230						13	1780	2500				
14	120	1600						14	2050	2960					14	1300						14	1910	2500				
15	150	1600						15	2440	2960					15	1400						15	2240	2500				
16	200	1600						16	2960						16	1400						16	2500					
17	250	1600						17	2960						17	1400						17	2500					
18	280	1600						18	2960						18	1400						18	2500					

TAL 2000 CLASS DIESEL LOCOMOTIVES

AIRBRAKE TRAINS >12.5 to 17 t/a								AIRBRAKE TRAINS >17 to 19 t/a							AIRBRAKE TRAINS >19 to 20 t/a							MOTOR CAR TRAINS, 8 to 12 t/a #						
CLASS : TAL 2000								CLASS : TAL 2000							CLASS : TAL 2000							CLASS : TAL 2000						
		(Number)							(Number)							(Number)							(Number)					
GC	GRAD.	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6
1	30	560	1120	1670	2230	-	-	1	570	1130	1700	2260	-	-	1	570	1140	1700	2270	-	-	1	410	830	1240	1660	-	-
2	35	660	1320	1980	2640			2	670	1340	2010	2680			2	670	1350	2020	2690			2	490	980	1480	1970		
3	40	760	1520	2280	3030			3	770	1550	2320	3090			3	780	1550	2330	3110			3	570	1130	1700	2270		
4	45	860	1710	2570	3400			4	870	1750	2620	3490			4	880	1760	2640	3520			4	640	1280	1920	2400		
5	50	950	1900	2860	3400			5	970	1950	2920	3800			5	980	1960	2940	3920			5	710	1420	2140	2400		
6	55	1050	2090	3140	3400			6	1070	2140	3210	3800			6	1080	2160	3240	4000			6	780	1560	2340	2400		
7	60	1140	2280	3400				7	1170	2340	3500	3800			7	1180	2350	3530	4000			7	850	1700	2400			
8	66	1250	2490	3400				8	1280	2560	3800				8	1290	2580	3880	4000			8	930	1860	2400			
9	70	1320	2640	3400				9	1360	2710	3800				9	1370	2740	4000				9	980	1960	2400			
10	80	1490	2980	3400				10	1540	3080	3800				10	1560	3110	4000				10	1110	2210	2400			
11	90	1660	3320	3400				11	1720	3440	3800				11	1740	3470	4000				11	1220	2400				
12	100	1820	3400					12	1890	3780	3800				12	1910	3830	4000				12	1340	2400				
13	110	1980	3400					13	2060	3800					13	2090	4000					13	1450	2400				
14	120	2130	3400					14	2220	3800					14	2250	4000					14	1550	2400				
15	150	2550	3400					15	2690	3800					15	2730	4000					15	1840	2400				
16	200	3170	3400					16	3380	3800					16	3450	4000					16	2240	2400				
17	250	3400						17	3800						17	4000						17	2400					
18	280	3400						18	3800						18	4000						18	2400					

RELEVANT TO VACUUM BRAKED TRAINS THE FOLLOWING RESTRICTIONS MAY NOT BE EXCEEDED:

a. Maximum train speed of 60 km/h.
b. Maximum average axle load per train of 18.5 ton per axle.
c. Maximum axle load per truck per axle of 20 ton.
d. Maximum train length of 40 trucks (160 axles).
e. Maximum train length on specific line as restricted in WTB.
f. Maximum axle load restriction on specific line.
g. Maximum total load of 2560 ton on category 1 lines where SASKOP-couplers are used.
h. Maximum number of locomotives in a consist is four (4).

Notes : a. 'GC' Means Gradient Category.

This document is classified as 'Confidential' and as such is authorized and approved for general distribution within Transnet Freight Rail. (The document and/or any information contained herein may not be disclosed outside of TFR without prior permission first being obtained from the Principal Engineer (Train Design Technologies))

5.22 GENERAL LOAD TABLE 'TIME SENSITIVE' LIGHT AIRBRAKE TRAINS

The enclosed load table indicates the maximum allowable tonnage that may be hauled by the corresponding number of locomotives on 'Time Sensitive' Light Airbrake Trains.

LOAD TABLES FOR LIGHT AIRBRAKE CONTAINER TRAINS

		CLASS : 6E (NUMBER)							CLASS : 8E1/16E/17E (NUMBER)							CLASS : 7E/10E (NUMBER)							CLASS : 18E (NUMBER)						
GC	GRAD.	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6
1	30	260	520	780	1040	1300	1560	1	290	580	860	1150	1440	1730	1	530	1050	1580	2110	-	-	1	300	630	970	1320	1670	2010
2	35	310	620	930	1240	1550	1860	2	340	680	1030	1370	1710	2050	2	620	1240	1870	2490			2	360	740	1150	1560	1970	2380
3	40	360	720	1080	1430	1790	2150	3	400	790	1190	1580	1980	2370	3	710	1430	2140	2850			3	410	860	1330	1800	2270	2740
4	45	410	810	1220	1620	2030	2440	4	450	890	1340	1780	2230	2680	4	800	1600	2410	3000			4	470	970	1500	2030	2550	3000
5	50	450	900	1350	1810	2280	2710	5	500	990	1490	1980	2480	2970	5	890	1770	2660	3000			5	520	1070	1660	2240	2830	3000
6	55	500	990	1490	1980	2480	2970	6	540	1090	1630	2170	2710	3000	6	970	1940	2910	3000			6	570	1170	1810	2460	3000	
7	60	540	1080	1610	2150	2690	3000	7	590	1180	1770	2360	2940	3000	7	1050	2100	3000				7	620	1270	1970	2660	3000	
8	66	590	1170	1760	2350	2940	3000	8	640	1280	1930	2570	3000		8	1140	2280	3000				8	670	1390	2140	2900	3000	
9	70	620	1240	1860	2470	3000		9	680	1350	2030	2700	3000		9	1200	2400	3000				9	710	1460	2250	3000		
10	80	690	1390	2080	2780	3000		10	760	1520	2270	3000			10	1340	2690	3000				10	790	1630	2520	3000		
11	90	770	1530	2300	3000			11	840	1670	2510	3000			11	1480	2950	3000				11	870	1800	2780	3000		
12	100	830	1670	2500	3000			12	910	1820	2720	3000			12	1600	3000					12	950	1950	3000			
13	110	900	1790	2690	3000			13	980	1950	2930	3000			13	1720	3000					13	1020	2100	3000			
14	120	960	1910	2870	3000			14	1040	2080	3000				14	1830	3000					14	1090	2240	3000			
15	150	1120	2230	3000				15	1210	2430	3000				15	2130	3000					15	1270	2810	3000			
16	200	1340	2670	3000				16	1450	2900	3000				16	2550	3000					16	1510	3000				
17	250	1510	3000					17	1640	3000					17	2870	3000					17	1710	3000				
18	280	1600	3000					18	1740	3000					18	3000						18	1810	3000				

		CLASS : 33D (Number)							CLASS : 34D (Number)							37D (Number)							38 CLASS (DIESEL / ELECTRIC MODE) (Number)						
GC	GRAD.	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6	GC	1	2	3	4	5	6
1	30	260	510	770	1030	1290	1540	1	340	680	1020	1360	1700	2040	1	380	760	1150	1530	1910	2290	1	320	640	950	1270	-	-
2	35	310	620	920	1230	1540	1850	2	400	810	1210	1620	2020	2430	2	460	910	1370	1820	2280	2730	2	370	750	1120	1500		
3	40	360	710	1070	1430	1780	2140	3	470	940	1400	1870	2340	2810	3	530	1050	1580	2110	2630	3000	3	430	860	1290	1720		
4	45	400	810	1210	1620	2020	2420	4	530	1060	1590	2120	2650	3000	4	600	1190	1790	2380	2980	3000	4	480	970	1450	1930		
5	50	450	900	1350	1800	2250	2700	5	590	1180	1760	2350	2940	3000	5	660	1320	1990	2650	3000		5	530	1070	1600	2140		
6	55	490	990	1480	1970	2470	2960	6	640	1290	1930	2580	3000		6	730	1450	2180	2900	3000		6	580	1170	1750	2330		
7	60	540	1070	1610	2140	2680	3000	7	700	1400	2100	2800	3000		7	790	1580	2360	3000			7	630	1260	1890	2520		
8	66	580	1170	1750	2340	2920	3000	8	760	1530	2290	3000			8	860	1720	2580	3000			8	690	1370	2060	2750		
9	70	620	1230	1850	2470	3000		9	800	1610	2410	3000			9	910	1810	2720	3000			9	720	1440	2170	2890		
10	80	690	1380	2080	2770	3000		10	900	1810	2710	3000			10	1020	2030	3000				10	810	1610	2420	3000		
11	90	760	1530	2290	3000			11	990	1990	2980	3000			11	1120	2240	3000				11	890	1780	2660	3000		
12	100	830	1660	2490	3000			12	1080	2160	3000				12	1220	2430	3000				12	960	1930	2890	3000		
13	110	890	1790	2680	3000			13	1160	2330	3000				13	1310	2620	3000				13	1030	2070	3000			
14	120	950	1910	2860	3000			14	1240	2480	3000				14	1400	2790	3000				14	1100	2200	3000			
15	150	1110	2230	3000				15	1450	2900	3000				15	1630	3000					15	1280	2560	3000			
16	200	1330	2670	3000				16	1730	3000					16	1950	3000					16	1530	3000				
17	250	1510	3000					17	1960	3000					17	2200	3000					17	1730	3000				
18	280	1600	3000					18	2070	3000					18	2330	3000					18	1830	3000				

GENERAL DEFINITION OF A LIGHT AIRBRAKE CONTAINER TRAIN

A Light Airbrake train is a train consisting of wagons with an average axle load for each vehicle not exceeding 18.5 tons and an average axle load for the train not exceeding 15 tons per axle, and is time sensitive by the nature of the traffic it contains

NOTE

Standard Load Tables as provided for airbrake trains previously within this document may also be utilised for Light Airbrake Container Trains. It merely needs to be noted that the table above is designed for 'Time Sensitive' traffic and using alternate Load Tables might result in a loss of time and a longer through-put time.

This document is classified as 'Confidential' and as such is authorized and approved for general distribution within Transnet Freight Rail. (The document and/or any information contained herein may not be disclosed outside of TFR without prior permission first being obtained from the Principal Engineer (Train Design Technologies))

5.23 GENERAL LOAD TABLES FOR PASSENGER TRAINS

The enclosed load tables indicate the maximum number of coaches to be hauled by the corresponding number of locomotives over the associated routes on Mainline Passenger Trains.

1. ROUTE : Johannesburg to Durban (Both Directions)

Section	Minimum number of Locomotives	Maximum number Coaches
1.1 Johannesburg – Durban/Danskraal	1 x 5E1	10
	2 x 5E1	17
	3 x 5E1	21
	1 x 6E1/17E	13
	1 x 18E	13
	2 x 6E1/17E	21
	2 x 18E	21

2. ROUTE : Johannesburg to Cape Town (Both Directions)

Section	Minimum number of Locomotives	Maximum number of Coaches
2.1 Johannesburg – Kimberley	1 x 5E1	13
	2 x 5E1	18
	3 x 5E1	21
	1 x 6E1/17E	16
	1 x 18E	16
	2 x 6E1/17E	21
	2 x 18E	21
2.2 Kimberley – De Aar	1 x 34D	16
	2 x 34D	21
2.3 De Aar – Beaufort West	1 x 7E	21
2.4 Beaufort West – Worcester	1 x 6E	11
	1 x 18E	11
	2 x 6E	21
	2 x 18E	21
2.5 Worcester – Cape Town	1 x 6E	11
	1 x 18E	11
	2 x 6E	21
	2 x 18E	21
	1 x 33D	10
	2 x 33D	21

This document is classified as 'Confidential' and as such is authorized and approved for general distribution within Transnet Freight Rail. (The document and/or any information contained herein may not be disclosed outside of TFR without prior permission first being obtained from the Principal Engineer (Train Design Technologies))

3. ROUTE : Worcester to Port Elizabeth (Both Directions)

Section	Minimum number of Locomotives	Maximum number of Coaches
3.1 Worcester – Oudtshoorn	1 x 33D 2 x 33D 1 x 35D 2 x 35D 3 x 35D	7 18 7 14 18
3.2 Oudtshoorn – Klipplaat	1 x 33D 2 x 33D 1 x 34D 2 x 34D	12 18 14 18
3.3 Klipplaat – Port Elizabeth	1 x 35D 2 x 35D 3 x 35D 1 x 34D 2 x 34D	7 14 18 9 18

4. ROUTE : Johannesburg to Port Elizabeth (Both Directions)

Section	Minimum number of Locomotives	Maximum number of Coaches
4.1 Johannesburg – Kroonstad/Gunhill/Bloemfontein	1 x 5E1 2 x 5E1 3 x 5E1 1 x 6E1/17E 1 x 18E 2 x 6E1/17E 2 x 18E	13 18 21 15 15 21 21
4.2 Bloemfontein – Noupoort/Springfontein	1 x 34D 2 x 34D 1 x 35D 2 x 35D 1 x 34D + 1 x 35D	19 21 14 21 21
4.3 Noupoort – Port Elizabeth	1 x 7E	21

5. ROUTE – Springfontein to East London (Both Directions)

Section	Minimum number of Locomotives	Maximum number of Coaches
5.1 Springfontein – East London	1 x 7E 2 x 7E	16 21

This document is classified as 'Confidential' and as such is authorized and approved for general distribution within Transnet Freight Rail. (The document and/or any information contained herein may not be disclosed outside of TFR without prior permission first being obtained from the Principal Engineer (Train Design Technologies))

6. ROUTE – Kimberley to Bloemfontein (Both Directions)

Section	Minimum number of Locomotives	Maximum number of Coaches
6.1 Kimberley - Bloemfontein	1 x 34D 1 x 35D 2 x 35D	21 15 21

7. ROUTE - Bloemfontein to Zastron (Both Directions)

Section	Minimum number of Locomotives	Maximum number of Coaches
7.1 Bloemfontein – Zastron / Thaba Nchu	1 x 35D 2 x 35D	14 18

8. ROUTE - Kroonstad/Gunhill to Danskraal/Ladysmith (Both Directions)

Section	Minimum number of Locomotives	Maximum number of Coaches
8.1 Kroonstad/Gunhill – Danskraal/Ladysmith	1 x 5E1 2 x 5E1 3 x 5E1 1 x 6E1/17E 1 x 18E 2 x 6E1/17E 2 x 18E	10 16 21 10 10 21 21

9. ROUTE – Johannesburg to Mafikeng (Both Directions)

Section	Minimum number of Locomotives	Maximum number of Coaches
9.1 Johannesburg – Mafikeng	1 x 34D 2 x 34D	12 18

10. ROUTE – Kimberley to Mafikeng via Vryburg(Both Directions)

Section	Minimum number of Locomotives	Maximum number of Coaches
10.1 Kimberley – Mafikeng	1 x 34D	17

This document is classified as 'Confidential' and as such is authorized and approved for general distribution within Transnet Freight Rail. (The document and/or any information contained herein may not be disclosed outside of TFR without prior permission first being obtained from the Principal Engineer (Train Design Technologies))

11. ROUTE – Johannesburg to Pretoria (Both Directions)

Section	Minimum number of Locomotives	Maximum number of Coaches
11.1 Johannesburg – Pretoria	1 x 5E1	11
	2 x 5E1	18
	3 x 5E1	21
	1 x 6E1/17E	15
	1 x 18E	15
	2 x 6E1/17E	21
	2 x 18E	21
	1 x 34D	15
	2 x 34D	21

12. ROUTE – Pretoria to Beitbridge/Messina (Both Directions)

Section	Minimum number of Locomotives	Maximum number of Coaches
12.1 Pretoria – Pyramid	1 x 5E1	11
	2 x 5E1	18
	3 x 5E1	21
	1 x 6E1/17E	15
	1 x 18E	15
	2 x 6E1/17E	21
	2 x 18E	21
12.2 Pyramid – Pietersburg	1 x 7E	21
12.3 Pietersburg-Groenbult/Messina/Beitbridge	1 x 34D	10
	2 x 34D	21

13. ROUTE – Pretoria to Komatipoort (Both Directions)

Section	Minimum number of Locomotives	Maximum number of Coaches
13.1 Pretoria – Komatipoort	1 x 5E1	11
	2 x 5E1	18
	3 x 5E1	21
	1 x 6E1/17E	14
	1 x 18E	14
	2 x 6E1/17E	21
	2 x 18E	21

This document is classified as 'Confidential' and as such is authorized and approved for general distribution within Transnet Freight Rail. (The document and/or any information contained herein may not be disclosed outside of TFR without prior permission first being obtained from the Principal Engineer (Train Design Technologies))

14. ROUTE – Kaapmuiden to Pietersburg via Hoedspruit and Groenbult (Both Directions)

Section	Minimum number of Locomotives	Maximum number of Coaches
14.1 Kaapmuiden – Hoedspruit/Phalaborwa	1 x 5E1	12
	2 x 5E1	18
	3 x 5E1	20
	1 x 6E1/17E	13
	1 x 18E	13
	2 x 6E1/17E	20
	2 x 18E	20
14.2 Hoedspruit – Groenbult/Pietersburg	1 x 34D	10
	2 x 34D	20

15. ROUTE – Krugersdorp/Waterval to Witbank (Via Ogies)

Section	Minimum number of Locomotives	Maximum number of Coaches
15.1 Krugersdorp - Witbank	1 x 5E1	11
	2 x 5E1	18
	3 x 5E1	21
	1 x 6E1/17E	14
	1 x 18E	14
	2 x 6E1/17E	21
	2 x 18E	21

16. ROUTE – Durban to Maputo (Both Directions)

Section	Minimum number of Locomotives	Maximum number of Coaches
16.1 Durban - Empangeni	1 x 5E1	11
	2 x 5E1	19
	1 x 6E1/17E	12
	1 x 18E	12
	2 x 6E1/17E	19
	2 x 18E	19
16.2 Empangeni – Golela/Maputo	1 x 34D	9
	2 x 34D	18

This document is classified as 'Confidential' and as such is authorized and approved for general distribution within Transnet Freight Rail. (The document and/or any information contained herein may not be disclosed outside of TFR without prior permission first being obtained from the Principal Engineer (Train Design Technologies))

17. ROUTE – Cape Town to Saldanha/Bitterfontein (Both Directions)

Section	Minimum number of Locomotives	Maximum number of Coaches
17.1 Cape Town - Kalbaskraal	1 x 35D 2 x 35D	14 20
17.2 Kalbaskraal - Saldanha	1 x 35D 2 x 35D	14 20
17.3 Kalbaskraal - Bitterfontein	1 x 35D 2 x 35D	12 20

18. ROUTE – Cape Town to Bredasdorp (Both Directions)

Section	Minimum number of Locomotives	Maximum number of Coaches
18.1 Cape Town - Bredasdorp	2 x 35D 3 x 35D	13 18

19. ROUTE – Paarl to Franschhoek (Both Directions)

Section	Minimum number of Locomotives	Maximum number of Coaches
3.19.1 Paarl - Franschhoek	1 x 35D 2 x 35D	10 18

20. ROUTE – Hermon to Porterville (Both Directions)

Section	Minimum number of Locomotives	Maximum number of Coaches
20.1 Hermon - Porterville	1 x 35D 2 x 35D	10 18

21. ROUTE – Wolseley to Prince Alfred Hamlet (Both Directions)

Section	Minimum number of Locomotives	Maximum number of Coaches
21.1 Wolseley - Prince Alfred Hamlet	1 x 35D 2 x 35D	9 16

22. ROUTE – Alicedale to Grahamstown (Both Directions)

Section	Minimum number of Locomotives	Maximum number of Coaches
22.1 Alicedale - Grahamstown	1 x 35D 2 x 35D	8 16

This document is classified as 'Confidential' and as such is authorized and approved for general distribution within Transnet Freight Rail. (The document and/or any information contained herein may not be disclosed outside of TFR without prior permission first being obtained from the Principal Engineer (Train Design Technologies))

23. ROUTE – Pietermaritzburg to Matatiele (Both Directions)

Section	Minimum number of Locomotives	Maximum number of Coaches
23.1 Pietermaritzburg - Matatiele	1 x 34D 2 x 34D	8 16

24. ROUTE – Kimberley to Ariamsvlei (Both Directions)

Section	Minimum number of Locomotives	Maximum number of Coaches
24.1 Kimberley - Ariamsvlei	1 x 33D/34D 2 x 33D/34D	12 20

25. ROUTE – De Aar to Noupoort (Both Directions)

Section	Minimum number of Locomotives	Maximum number of Coaches
25.1 De Aar to Noupoort	1 x 7E	21

This document is classified as 'Confidential' and as such is authorized and approved for general distribution within Transnet Freight Rail. (The document and/or any information contained herein may not be disclosed outside of TFR without prior permission first being obtained from the Principal Engineer (Train Design Technologies))